GlucoEdge: Quantization-Calibration Coverage and Staged Model Selection in On-Device Glucose Trend Classification

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Abstract

GlucoEdge, an Android/iOS research prototype, classifies 12 five-minute continuous glucose monitor (CGM) readings into five classes of the mean rate of change over the next 15 minutes. We ask how quantization-calibration coverage affects directional (minority-class) recall, and which changes improve it in staged, pre-registered sweeps. For a 2,909-parameter convolutional network in 8-bit integers (INT8), range stratification, the primary contrast, was null; random selections already covered the input range. In a prespecified secondary contrast, contiguous first-N selection, capping inputs at 250.9 mg/dL, cost 13.4 points of held-out directional recall (95% CI 10.3–16.9) against uniform random selection; a clip-only control attributes the loss to clipping; the fix is free to deploy. Two small cohorts did not confirm frozen predictions; post hoc, difference-encoded inputs (variant chosen on held-out data) cut the effect to 0.5 points. Rounds one to three reused one Weinstock test split. Round one's macro-averaged F1 (macro-F1) winner passed every gate but was rejected post hoc: held-out INT8 directional recall fell (0.517 to 0.438). Round five's seed-4 artifact (flatten head, cosine schedule, validation-chosen epoch), now bundled in the research applications, passed all gates on a never-evaluated 2008 trial cohort by +0.20 points [+0.11, +0.29], against round three's 1.1-point seed spread, but lowered macro-F1 there (0.410 vs. 0.427) and failed two gates on both Weinstock components. GlucoEdge is not a medical device.

1 Introduction

A continuous glucose monitor (CGM) sampled every five minutes produces 12 readings per hour, from which a near-future glucose value, event, or trend can be predicted [1]. This paper studies such predictions for research purposes, not to guide treatment.

These targets define different learning problems (value regression, event prediction, trend classification), and studies further differ in horizon, contextual inputs, cohort, split construction, and metric. Reviews, comparison indices, public benchmarks, and cross-dataset experiments document these differences and help separate within-dataset performance from transfer [1–4], so scores from incompatible protocols are not directly comparable.

Commercial CGM trend arrows are related but not equivalent: they summarize recent or current direction and rate with device-specific rate bands and look-back intervals [5, 6], whereas GlucoEdge labels each window by the mean rate of change over the next 15 minutes, from the current sample and the sample 15 minutes later — a research representation that neither replicates a manufacturer's algorithm nor gives a dosing rule.

Deployment raises further questions: conversion may change a model's outputs, the client may need different input handling, and lower precision does not guarantee lower latency on a given device. Class imbalance can hide conversion errors, because accuracy can rise as predictions concentrate on a majority class while minority-class recall declines [7].

This paper asks two research questions. The first is how quantization-calibration coverage affects the minority-class recall and deployment cost of a compact glucose trend classifier. Static 8-bit integer (INT8) post-training quantization (PTQ) sets activation ranges from a set of calibration inputs [8, 9], and coverage means the input values and trajectories that set represents. Unqualified, calibration here means this quantization step, neither CGM sensor calibration against blood glucose nor calibration of predicted probabilities. The second question is which model and training-data changes improve directional recall, the mean recall (sensitivity) of the four non-stable classes, when model selection runs in staged rounds that may each spend at most one pre-registered confirmatory read of protected labels.

Representative calibration data is established guidance: converter documentation asks for a few hundred samples "randomly chosen, in no particular order" [10]. This work measures, with participant-clustered intervals and a clip-only control, how contiguous first-N (dataset-order) selection of overlapping, time-ordered windows violates that guidance, and the class-level consequence: the clipped inputs concentrate in the fast-falling class, whose recall collapses while accuracy rises (Sections 7.4

and 9.1). The question arose in this project's own pipeline, whose conversion script calibrated the originally bundled INT8 artifact on the first 200 validation windows: 17.5 hours of one participant's readings, whose maximum lies below readings found in a quarter of training windows (Section 7.1). Earlier revisions scored the same held-out participants under an uncorrected protocol (Appendix A), so the study is a locked, corrected reanalysis rather than a fresh test.

The calibration study holds the architecture, loss, quantizer, and data split fixed and varies only calibration selection: four strategies (sequential, the contiguous first-N selection; uniform random; participant-balanced random; participant-balanced range coverage), two nested budgets (200 and 1,000 windows), three random draws, and five training seeds, giving 100 INT8 artifacts evaluated against their five float sources, plus a ten-artifact min–max-observer arm. Its hypotheses, samplers, budgets, endpoint, bootstrap, and missing-class rules were frozen before any held-out label was read (Table 16, Appendix C).

The second question is addressed by five rounds of staged greedy selection (Section 6.2) that train on Weinstock or, from round two on, on cohorts of the public DiaData collection [11] (Table 11, Appendix B), select on Weinstock validation windows, and decide each candidate with frozen gates. Rounds one to three confirmed on the Weinstock test components that the calibration study had already read, and round five on RT-CGM, a randomized-trial cohort that no round had evaluated (Table 13, Appendix B). This paper reports:

- a corrected, provenance-masked evaluation protocol for GlucoBench-formatted CGM windows, with eligible-window and class-support audits published alongside the code;
- a controlled answer to the coverage question (Section 7.3): in a prespecified secondary contrast, contiguous first-N selection lost 13.4 points of held-out directional recall (95% confidence interval (CI) 10.3–16.9) against uniform random selection of the same size, which restored float-level recall. The prespecified primary contrast, range coverage versus participant-balanced random, was null (+0.06 points, CI −0.02 to +0.13) and could not probe range coverage, because both arms received the same input range. Where every randomized selection covers the full input range, as in this pool, stratification, a five-fold budget, and the observer choice move recall by at most about a tenth of a point;
- mechanism controls that attribute the loss to input-range clipping rather than input-grid resolution or internal quantization: the clip-only input image, fed to the float model, reproduces 13.14 points of the sequential arm's 13.35-point float-to-INT8 loss (Section 7.4);
- deployment-cost accounting: every INT8 artifact of the 2,909-parameter study model has the same size and operator inventory, selection takes under 0.03 s, and a prespecified nine-artifact iPhone benchmark found the calibration arms within 0.05 μs of each other, so the corrected calibration is free at deployment. INT8 showed no mean-latency advantage over float in any measurement and saves 31% of file size for the study model and 70% for the bundled one;
- two exploratory follow-ups: on two small further cohorts the frozen predictions were not confirmed (Dubosson +2.0 points, exact sign-flip $p = 0.094$; Hall inconclusive, its premise failed; Table 8), and in a post-hoc experiment difference-encoded inputs shrank the uniform-versus-sequential difference to +0.5 points (Table 18, Appendix C); and
- five staged accuracy sweeps (Sections 8 and 9.3); four spent one pre-registered confirmatory read each and round four, whose candidate converted to the champion's own artifacts, none: a macro-averaged F1 (macro-F1) winner that passed every gate but was rejected post hoc because held-out INT8 directional recall fell from 0.517 to 0.438; DiaData-only training, bundled after round two; a label-definition stability check; and the seed-4 flatten-head artifact (cosine schedule [12], validation-chosen epoch), now bundled in the research applications, which passed all gates on never-evaluated RT-CGM by +0.20 points (CI +0.11 to +0.29), a fifth of round three's 1.1-point float seed spread on Weinstock held-out (its own RT-CGM seed spread was not measured), but failed two gates on both Weinstock components and lowered RT-CGM macro-F1 (0.410 against 0.427); PEDAP is a near-tie (0.539 against 0.541), and its four changed factors are not separately identified.

Source code, tests, synthetic replay data, and the bundled model artifacts (released for non-commercial research use only) are in the project repository.[1]

2 Related Work

2.1 Forecasting tasks, horizons, and inputs

Forecasting from CGM has expanded from autoregressive value prediction on sensor history [13] to recurrent and attention models with contextual inputs [14], recurrent variance estimation [15], and convolutional–recurrent fusion of CGM, insulin, and meals [16]. Architecture and input modality change together across these studies, so their results do not isolate model family; cross-dataset evaluation asks instead whether a trained model survives a change of dataset, cohort, and collection process [4]. Table 1 summarizes these design differences, sep-

[1] `https://github.com/mohamedmenasy/GlucoEdge`; release v1.1.0 is archived on Zenodo, `https://doi.org/10.5281/zenodo.23426171`.

arating offline predictive evaluation from reported device execution.

2.2 Benchmarks and evaluation protocols

The split determines what a study can claim about generalization: subject-dependent splits evaluate people represented during training, held-out-participant splits evaluate unseen people in the same dataset, and cross-dataset evaluation also changes the cohort and pipeline. GlucoBench standardizes preprocessing, splits, and baselines across five public datasets [1]; experiments across four datasets directly test changes in data source [4]. The calibration study uses Weinstock, GlucoBench's largest type-1 diabetes cohort [17]; the accuracy sweeps also train on five DiaData cohorts [11] and evaluate zero-shot, with models that never trained on them, on PEDAP (Pediatric Artificial Pancreas trial) [18] and RT-CGM, the Juvenile Diabetes Research Foundation (JDRF) CGM randomized trial [19] (Table 11).

Nearby and overlapping windows are dependent, so time-series splitting requires care: cross-validation validity for autoregressive prediction depends on the residual structure [20], leakage can answer a less demanding question than the one claimed [21], and even with a fixed split, training and implementation choices contribute benchmark variance [22]. Results therefore need to state their participant, temporal, and dataset boundaries.

2.3 Trend representations and class imbalance

Most Weinstock labels are stable and faster changes are rare, so overall accuracy weights the stable class by its prevalence. Macro-averaged recall (balanced accuracy in this single-label setting) weights each class equally and makes minority-class degradation visible [7], though it does not establish event-level or clinical performance.

No published baseline exists for this exact task (five rate-of-change classes at a future sample, scored on held-out participants). The nearest classification study predicts three glucose *level* bands (hypo-, eu- and hyperglycemia) for 11 participants whose data were recorded every 15 minutes, so its 15-minute horizon is one step ahead; its best models reach recalls of 0.85–0.98, with euglycemia recall of 0.59–0.70 at 1 h, under within-patient splits [25]. Level bands inherit the level's autocorrelation, three classes are coarser than five rate bands, and within-patient splits exploit person-specific dynamics that a held-out protocol denies.

Rate and direction are already graded in this domain. Continuous glucose-error grid analysis (CG-EGA) pairs a point-accuracy grid with a rate-error grid [26] and has been applied to glucose predictors [27]; the J index combines a forecast's regularity with its time gain [3]; and coherence penalties discourage predicted changes that contradict the measured ones [28]. Because the target here is a class rather than a value, this paper reports per-class recall (sensitivity) and its directional mean, which expose the minority-class losses that accuracy hides. Per-class recall counts every misclassification alike: it does not distinguish an adjacent rate band from a reversed direction, weight errors by glucose level, or grade timing, all of which CG-EGA and the J index address.

2.4 Edge inference and quantization

One-dimensional convolutional neural networks (CNNs) extract local temporal features with compact feed-forward networks [29]. Mobile inference systems can reduce execution cost through compression and runtime-specific scheduling [30], but latency is a property of model, runtime, and hardware together, so smartphone and tiny-device benchmarks measure specified workloads on specified platforms [31, 32]. Glucose predictors have been run on such hardware: Li et al. timed a convolutional–recurrent forecaster on Android [16], and Zhu et al. embedded a recurrent predictor in a wearable whose low-power system-on-chip performs real-time glucose prediction and hypoglycemia detection [24]. Both forecast values, not trend classes (Table 1).

Integer quantization approximates floating-point tensors with discrete values parameterized by a scale and zero point [8]; static PTQ estimates activation ranges from calibration inputs before conversion [9, 10]. Low-bit studies show that clipping, range estimation, and rounding choices can determine the error [33, 34], and that accurate PTQ is attainable from small calibration sets [35].

Which samples calibrate the quantizer is itself a studied lever. Standard INT8 recipes set activation ranges from a small calibration set [36, 37], and range estimators (maximum, entropy, percentile) have been compared across vision, speech, and language networks [38]. In language models, the choice of calibration data alone produces substantial variation in downstream performance after quantization or pruning [39]; in vision, PTQ accuracy tracks how closely the calibration set's feature statistics match the source data's, more than whether it comes from the same domain [40]; and data-free methods synthesize calibration inputs that match a network's batch-normalization statistics [41]. Converter documentation states the practical rule: the representative dataset is "a small subset of a few hundred samples randomly chosen, in no particular order, from the training or evaluation dataset" [10], and the LiteRT (formerly TensorFlow Lite) guide asks for around 100–500 training or validation samples [42]. That calibration data must represent the deployment inputs is therefore established guidance, not a finding of this paper.

Table 1: Representative CGM forecasting studies and evaluation scope. Metrics are not compared across rows because datasets and protocols differ.

Study	Target	Horizon	Inputs	Evaluation scope	Device evidence
Reifman et al. [13]	Value	30 min	CGM	9 participants	None reported
Mirshekarian et al. [14]	Value	30/60 min	CGM + context	OhioT1DM [23]/synthetic	None reported
Martinsson et al. [15]	Value + variance	30/60 min	CGM	OhioT1DM [23]	None reported
Li et al. [16]	Value	30/60 min	CGM, insulin, meals	10 real + 10 simulated	Android timing
Zhu et al. [24]	Value + hypoglycemia	30/60 min	CGM + time of day	OhioT1DM [23] + two private datasets (47 participants); in silico trial	Wearable system-on-chip
GlucoBench [1]	Value trajectory	60 min	Dataset-dependent	Five public datasets	None reported
Ghimire et al. [4]	Value	30/60 min	CGM	Four datasets	None reported
GlucoEdge (this work)	Trend class	15 min	CGM	Training on Weinstock or five DiaData cohorts; Weinstock held-out and temporal, RT-CGM and PEDAP zero-shot	Android/iOS replay timing

3 Problem Formulation

Let g_t be the CGM sensor reading at time t, in mg/dL, with observations every five minutes. The input is

$$\mathbf{x}_t = [g_{t-55}, g_{t-50}, \ldots, g_t] \in \mathbb{R}^{12}. \tag{1}$$

The target is the mean rate of change over the next 15 minutes, computed from a point outside the input window:

$$r_t = \frac{g_{t+15} - g_t}{15} \quad [\text{mg/dL/min}], \tag{2}$$

not the instantaneous rate at $t + 15$. The class y_t is assigned by

$$y_t = \begin{cases} \texttt{falling_fast}, & r_t \le -2, \\ \texttt{falling}, & -2 < r_t \le -1, \\ \texttt{stable}, & -1 < r_t < 1, \\ \texttt{rising}, & 1 \le r_t < 2, \\ \texttt{rising_fast}, & r_t \ge 2. \end{cases} \tag{3}$$

The 12 readings span 55 elapsed minutes ("one-hour history" means 12 samples at a five-minute cadence); the history length and 15-minute horizon are fixed engineering choices for this prototype. The five-class target, a simplification based on two samples, does not describe the full intervening trajectory. The boundaries are trend-arrow rate bands (± 1 and ± 2 mg/dL/min), the rates that CGM trend arrows commonly separate; the five classes collapse device-specific arrow taxonomies [5, 6]. No threshold was optimized on the test labels.

4 Data and Methods

4.1 Dataset and preprocessing

The experiments use the Weinstock data distributed through GlucoBench. The source study was a case-control study of severe hypoglycemia in adults aged 60 or older with type-1 diabetes for at least 20 years: 101 cases had at least one severe hypoglycemic event in the prior year, 100 age-matched controls had none in the prior three years, and cases showed greater glucose variability [17]. The cohort is therefore enriched for hypoglycemia and variability relative to older adults with type-1 diabetes in general, which can shift the class prevalences below and limits how far the results transfer. CGM data were collected blinded, so participants could not react to their readings: the trajectories lack the corrections that a real-time display or closed-loop control would prompt, and deployed use may see different dynamics. GlucoBench lists the sensor as a Dexcom G4 and reports 200 participants before preprocessing and 192 afterwards, at a five-minute sampling interval [1]; an audit of the formatter output reproduces both counts. Scaling is disabled, so glucose stays in raw mg/dL. The formatter rounds timestamps to the minute, splits each participant's record at gaps longer than 45 minutes, drops segments with fewer than 240 original readings, linearly interpolates each remaining segment onto a five-minute grid that starts at its first reading, and tags it with an `id_segment`. Of the 579,782 grid values, 75.8% coincide with an original reading, 19.8% fall between two consecutive readings no more than 7.5 minutes apart (reading times drift relative to the grid), and 4.4% fill one or more missing readings. Because interpolation precedes windowing, interpolated values may occur among the inputs or at target samples. The configured split holds out 10% of participants (20 people) as an out-of-distribution (held-out-participant) component and forms temporal validation and test components from the rest; the formatter combines the test components into one frame, which Section 4.2 separates again by row ownership. All results report the held-out-participant and temporal components separately.

The readings span exactly 39–401 mg/dL. Dexcom

Table 2: Window distribution after GlucoBench formatting. Percentages are within each split.

Class	Train	Aggregate test
`falling_fast`	13,260 (3.12%)	3,692 (2.90%)
`falling`	40,122 (9.45%)	11,387 (8.95%)
`stable`	314,257 (74.04%)	95,998 (75.49%)
`rising`	39,428 (9.29%)	11,012 (8.66%)
`rising_fast`	17,395 (4.10%)	5,076 (3.99%)
Total	424,462	127,165

G4 systems report 40 to 400 mg/dL and display LOW or HIGH outside that range, and the data pile up at the two values just beyond it: among the distinct readings in eligible held-out windows (Section 4.2), 922 equal 39 mg/dL and 1,253 equal 401 mg/dL, against 33 at 40 and 22 at 400. The two values are therefore consistent with stored LOW and HIGH codes (the source data dictionary was not checked). A code bounds the true value rather than measuring it, so a window whose current or endpoint value is coded has a label rate that is itself only a bound. Of the eligible held-out windows, 6.6% contain a coded input and 4.2% a coded endpoint (temporal: 4.7% and 2.9%); by true class, the share with a coded input runs from 3.3% (`falling`) to 9.8% (`falling_fast`), and coded endpoints concentrate in `stable` (5.4%, against 0.7–1.7% in the directional classes). Table 17 (Appendix C) includes a check that excludes them.

Windows are generated independently within each (`id`, `id_segment`) group, in time order, and emitted only when all 12 input points and the three following horizon steps fit in the same segment, so a sensor gap or participant boundary is never treated as a continuous trajectory. The stride is one reading, matching an application that updates every five minutes; adjacent windows therefore overlap heavily and are statistically dependent. The frames yielded 424,462 training, 55,289 validation, and 127,165 aggregate test windows from 172, 127, and 186 participants: 57,068 test windows from the 20 held-out participants and 70,097 from temporal segments of 166 training participants.

Table 2 shows the resulting class imbalance; both fast classes exceed 13,000 training windows. A preliminary 20-epoch run obtained test recalls of 0.5333 (`falling_fast`) and 0.5380 (`rising_fast`), so neither met the predeclared condition for collapsing the classes (fewer than 200 training examples or recall below 0.30), and the classifier keeps all five. Test recall informed this five-versus-three decision, an exploratory development choice rather than independent validation.

4.2 Split roles, observation mask, and eligible windows

GlucoBench repeats rows from a preceding split at the top of validation and in-distribution test frames so sequence models can form initial histories. Earlier revisions let every prefix row start a window, so 13.6% of aggregate test windows scored targets in validation-owned rows and 35.1% of validation windows duplicated training windows (Table 14, Appendix C). The corrected protocol assigns every formatter row to the earliest split that lists it and scores a window only when its anchor (last input sample) and all three forecast rows are owned by that split; earlier-split readings may serve as causal input history, and scored (participant, segment, target time) keys are asserted disjoint across splits.

Table 3: Eligible windows under the corrected protocol. Scored removes context-only and boundary-crossing targets; eligible also removes windows with gap filling.

Component	Frame	Scored	Eligible	Part.
Training	424,462	424,462	328,575	172
Calibration pool	55,289	35,560	27,109	127
Temporal test	127,165	52,540	37,361	166
Held-out test		57,068	46,376	20

Because the formatter interpolates before windowing, a second mask removes gap filling: a grid value is retained only if it lies on or between two original readings at most 7.5 minutes apart. An audit showed that requiring exact coincidence with an original reading would discard roughly 40% of windows for grid drift unrelated to missing data, so the gap-fill rule is the primary mask and exact coincidence a sensitivity tier. A window is eligible when its 12 inputs and future endpoint are all gap-free; the two intermediate forecast steps are not label inputs and are not masked. Table 3 shows the result: eligible training windows retain all five classes with at least 10,330 examples (`falling_fast`), and the calibration pool keeps 127 of 172 development participants.

4.3 Model and training

The calibration study uses `TrendCNN`, a one-dimensional CNN on tensors of shape $(B, 1, 12)$: a convolution from 1 to 24 channels, a second from 24 to 36 channels (both with kernel size 3, stride 1, and padding 1), a rectified linear unit (ReLU) after each, adaptive average pooling over time, and a 36-to-5 linear head, 2,909 trainable parameters in all. Later experiments mostly keep this family (round three also tried a temporal convolutional network [43]) and vary its input encoding, width, pooling head, loss, and training data; the models bundled in the applications over the project's history range from 2,909 to 50,645 parameters and were trained on Weinstock or on DiaData cohorts [11]. Table 12 (Appendix B) lists every bundled variant with its architecture, encoding, loss, training data, quantization-calibration pool, and artifact hash; Figure 1 shows the shared workflow.

Training uses inverse-frequency class-weighted cross-entropy, batch size 64, and Adam with learning rate 10^{-3} for 20 epochs [44, 45] in the calibration study and its follow-up experiments (Section 6.2 gives the sweeps' settings). The study trains five checkpoints, seeds 0–4, on

the eligible windows of Section 4.2; the fixed split does not change with the training seed. Validation loss was observed after every epoch, with no early stopping or test-time selection. Metrics were computed with scikit-learn and the study's own confusion-count code [46]. Because the test distribution is dominated by `stable`, the primary summary is the unweighted mean recall (per-class sensitivity) of the four directional classes (directional recall); five-class macro recall, equivalent to balanced accuracy, is also reported [7].

Two input-dependent baselines use the same eligible windows: a recent-slope baseline that maps $(g_t - g_{t-15})/15$, from the inputs alone, through the same five thresholds, and a multinomial logistic regression on the 12 readings with training-only standardization, balanced class weights, and fixed $C = 1$ (converged). A constant-`stable` classifier shows the effect of prevalence.

5 Conversion and Deployment

5.1 Conversion and quantizer

Every checkpoint is converted with LiteRT Torch, the PyTorch converter for LiteRT (Google's on-device inference runtime, formerly TensorFlow Lite): float conversion with a fixed $(1, 1, 12)$ input signature, and static INT8 PTQ through PyTorch 2 Export (PT2E) with LiteRT Torch's `get_symmetric_quantization_config` helper (`is_per_channel=True`, `is_dynamic=False`). Despite the helper's name, only the weights are quantized symmetrically, per channel; activations, the model input included, use per-tensor affine quantization whose ranges a histogram observer selects to minimize quantization error [9]. The observer always includes zero, so with zero point -128 an input range runs from 0 to 255 times the input scale, and the codes below the lowest CGM reading go unused. The project's conversion script originally took its 200 calibration windows from the start of the validation set in dataset order, with no randomization or range balancing (Section 7.1). In the converted graph only the final fully connected layer executes in int8; both convolutions run in float32 on dequantized tensors (Section 7.9).

5.2 Calibration experiment design

The study freezes the checkpoint, quantizer, and conversion pipeline and varies only which windows of the calibration pool, the 27,109 eligible validation windows, calibrate the histogram observer; no sampler reads future labels, and selection code is tested to be invariant under label shuffling. Four strategies are compared. *Sequential* is contiguous first-N (dataset-order) selection: the first N windows in deterministic (participant, segment, time) order, as the project's original conversion script took them; the guidance this pipeline cites asks for random selection (Section 9.1). *Uniform random* is a seeded permutation of the pool. *Participant-balanced random* cycles a per-draw shuffled participant order round-robin, taking one window from a fixed per-participant permutation each visit and skipping exhausted participants. *Participant-balanced range coverage* reuses exactly that participant allocation but, within each participant, cycles through low/middle/high range cells defined by the pool's own 10th and 90th percentiles of per-window input minimum and maximum, so it differs from balanced random only in within-participant input-range coverage.

Each strategy is drawn at budgets of 200 and 1,000 windows, the 200-window set nested as the first 200 of the 1,000-window ordering. Stochastic strategies use three seeded draws (101–103); every selection is shared across the five model seeds, giving 100 INT8 artifacts (5 seeds $\times$ [2 sequential + 3 strategies $\times$ 2 budgets $\times$ 3 draws]). A mechanism-control arm re-quantizes two cells (sequential-200 and range-coverage-200, draw 101) with a min–max activation observer in place of the histogram observer, adding 10 artifacts. For every calibration set the study records participants, segments, unique input readings, input extremes and quantiles, and occupied range cells, because a nominal window count can cover few distinct readings; for every artifact it records input quantization parameters, size, and SHA-256.

5.3 Client implementations

For input scale s and zero point z, both applications use the affine form $q = \mathrm{clip}(\mathrm{round}(x/s) + z, -128, 127)$, $\hat{x} = s(q - z)$, with quantization parameters read from the model metadata rather than hardcoded. All three implementations divide in single precision and round exact half-way values to the even integer — Python with `numpy.round`, Kotlin with `kotlin.math.round`, Swift with `rounded(.toNearestOrEven)` [47–49] — so they produce identical integer inputs for any glucose value.

The Kotlin/Jetpack Compose application and its SwiftUI iOS mirror replay a bundled synthetic trace through a 12-reading ring buffer that resets after a cadence gap, mirroring training-time segment boundaries, and classify every complete window locally with the float artifact by default or INT8 for comparison. Both run byte-identical models through LiteRT's `CompiledModel` API [50] (LiteRT 2.1.0 on Android, the 2.3.0 release binaries on iOS); on real devices both use its CPU accelerator, which executes the graph through the XNNPACK delegate, with no further options; neither sets a thread count. The merged Android manifest is tested to ensure it does not request the `INTERNET` permission; iOS has no equivalent, so a source scan fails the continuous-integration build if networking APIs appear in the application code, a weaker guard. Both applications also offer an optional on-device caption of the replayed window ("Explain"; Apple Foundation Models on iOS, LiteRT-LM on Android), instructed never to give advice, recommendations or dosing and labelled "On-device demo note — not medical guidance"; the captions were not evaluated. Appendix C describes the iOS packaging and an emulator workaround.

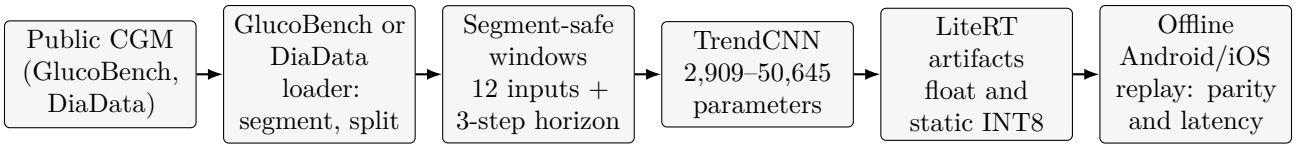


Figure 1: End-to-end GlucoEdge workflow. The calibration study uses Weinstock data and the 2,909-parameter TrendCNN; the accuracy sweeps add DiaData cohorts and wider variants (Table 12). Patient data are used locally for training and evaluation; only model files and a synthetic replay trace are committed to the applications.

6 Experimental Protocol

6.1 Calibration study protocol

Evaluation uses two disjoint components from Section 4.2, never concatenated: the 46,376 eligible windows of the 20 held-out participants (primary) and the 37,361 eligible temporal windows of 166 training participants (secondary). The protocol (samplers, budgets, endpoint, bootstrap, and missing-class rules) was frozen and committed before any held-out label was read, in `paper/results/calibration-coverage/protocol.json`; its hypotheses H1–H4 are stated in the study plan that file names, committed at the freeze commit it records (`d63cc8f`). Table 16 (Appendix C) maps each hypothesis and contrast to its outcome. A provenance audit (`experiments/split_audit.py`) maps every window to the formatter rows it uses (Section 4.2), reading row identities and timestamps for all splits but glucose values only from training and validation rows; it also records the originally bundled calibration set and that artifact's input quantization parameters.

With class support n_c, $N = \sum_c n_c$, and class recall R_c, accuracy is $A = \sum_{c=1}^{5}(n_c/N)R_c$ and macro recall $R_{\text{macro}} = \frac{1}{5}\sum_{c=1}^{5} R_c$; the primary endpoint is the unweighted mean recall of the four directional classes. The prespecified primary contrast is range-coverage versus participant-balanced random at budget 200, which was designed to isolate within-participant input-range coverage from participant representation. Secondary contrasts are range coverage versus uniform random, 1,000 versus 200 within each strategy, the budget-by-strategy interaction, and uniform random versus sequential at each budget, the last added by a protocol amendment before any held-out label was read; the study plan marks secondary intervals descriptive. `protocol.json` names the interaction without a formula, so Section 7.3 reports the difference between two strategies' 1,000-versus-200 effects, a definition fixed after the primary analysis was published. A recall is undefined on zero support and never set to zero.

Uncertainty comes from a paired participant-cluster bootstrap: 2,000 draws with a fixed analysis seed resample participants with replacement, keep every selected participant's complete windows, recompute each arm's confusion counts, and average the paired differences over the five model seeds and three calibration draws inside each bootstrap draw. Percentile 95% CIs are conditional on the fitted models and calibration sets; model-seed and calibration-draw variation are reported separately. Overlapping windows are never resampled individually. Draws with an undefined endpoint are dropped before the percentile step, and if more than 5% of draws are undefined, the interval is omitted.

Two input-grid controls separate mechanisms. The float quantize–dequantize (QDQ) control feeds the float model each INT8 artifact's QDQ image of the inputs, with that artifact's own scale and zero point, which measures the input-grid intervention alone; the clip-only control clips inputs to the artifact's representable range without re-gridding, which separates range truncation from input resolution. Clipping is counted with the actual quantization operation (round half-to-even, then saturate), per input slot and per window, with true-class breakdowns on the evaluation components only. These comparisons describe the measured error components, not an additive causal decomposition of every error.

Two follow-up experiments reuse this machinery under their own frozen protocols. A second-cohort run repeats a reduced matrix on two further GlucoBench cohorts, Dubosson [51] and Hall [52]: five seeds, sequential versus uniform random at budget 200, and one directional prediction per cohort, frozen before that cohort's evaluation labels were read (Table 8). A delta-encoding run, a post-hoc experiment on the already-read Weinstock held-out data, retrains the model on re-encoded inputs under a frozen float gate, robustness prediction, and ship gate (Table 18).

File size is the exact serialized artifact size. A prespecified iPhone benchmark times nine artifacts, the seed-0 float artifact and 8 of the 110 study INT8 artifacts; the other 102 study INT8 artifacts and all second-cohort and delta-encoded artifacts were not timed. Cross-runtime parity tests run 20 synthetic golden windows through Python, Kotlin, and Swift and verify each bundled artifact's SHA-256 hash. Appendix C gives both protocols.

6.2 Accuracy sweep protocol

The second research question asks which model and training-data changes raise the directional recall of the bundled INT8 model. Five rounds of staged greedy selection addressed it (Section 8): each stage trains a few arms that differ in one factor (two in round one's width-by-window grid and round five's recipe arm), compares their seed means, and carries the winner forward,

and a final stage picks one seed. Selection reads only the 27,109 eligible Weinstock validation windows of Section 4.2, whatever the training data, with three seeds per arm in rounds one and two (five at their final seed stage) and five from round three on. Arms within 0.1 point of each other are tied and the cheaper arm wins (fewer parameters first); Appendix D records how rounds three and five applied this rule.

Round one selected on validation macro-F1 and confirmed on held-out macro-F1. Its outcome (Section 8.1) changed the endpoint: from round two on, selection and confirmation use directional recall, guarded by a stable-recall floor that disqualifies an arm at selection if its mean validation stable recall falls more than 5 points (rounds two to four) or 2 points (round five) below the champion's. The champion is the pair bundled when a round began: the delta-encoded pair (Table 18) in rounds one and two and the round-two winner in rounds three to five (Table 12).

From round two on, three gates decide a candidate: (1) its INT8 model beats the champion's INT8 model on the endpoint, with the 95% interval's lower bound above -0.5 points; (2) its INT8 stable recall is at most 2 points below the champion's; (3) its INT8 directional recall is within 0.5 points of its own float. Round one applied gate 1 to macro-F1, used the float-proximity rule as its second gate, and reserved its third for ensembles. A round spends at most one confirmatory read: its protocol file (`protocol_sweep_r`N`.json`) is committed before any protected label is read, and the evaluation code refuses to run until it is. Rounds one to three read the Weinstock held-out (primary) and temporal (secondary) components, and from round two PEDAP as a descriptive probe. Round five moved its primary endpoint to RT-CGM, a cohort no round had evaluated, and declared the Weinstock components and PEDAP non-gating secondaries. A candidate whose converted artifacts are SHA-256-identical to the champion's spends no read (round four's outcome, made a rule in round five), so four rounds spent one read each and round four none; Table 13 lists every read of held-out, temporal and RT-CGM labels in this work. No protocol stated when a gate-passing candidate may be overridden. The two cases that arose fit one rule: override only if the candidate loses directional recall on the primary component (round one), not for losses on components the protocol declared non-gating (round five).

Round one trained on the eligible Weinstock training windows, its pooling stage adding the Dubosson and Hall training windows. Rounds two to five drew on DiaData v3 [11], using only its raw (non-imputed) five-minute archive. An overlap and sensor audit (`diadata_audit_r1.json`) admitted six type 1 diabetes (T1D) cohorts with 1,001 subjects, excluding DiaData's copies of the Weinstock and Dubosson studies and cohorts with a non-five-minute or unrecorded sensor interval, RT-CGM among them. Of the six, five (CITY, DCLP3, REPLACE-BG, SENCE and WISDM; 898 subjects) fed training, and PEDAP (103 subjects) trained no model. Mixtures were capped near 15 million training windows per arm by seeded whole-subject sampling, so no subject is split. Round three reopened RT-CGM, DiaData's label for the JDRF CGM randomized trial, for one training arm, and round five used it as its primary endpoint; Table 11 characterizes every cohort.

All sweep models take the delta-anchor input of Table 18 (11 consecutive differences and the last reading divided by 16) over a 12-reading window, except arms that change the input. Training uses Adam with learning rate 10^{-3} for 20 epochs, keeps the final epoch, and uses the study's inverse-frequency class-weighted cross-entropy (CE) unless a stage varies the loss. Round one trained with batch size 64 on the CPU, as the calibration study did; rounds two to five trained with batch size 512 on an Apple-silicon graphics processor through Metal Performance Shaders (MPS), chosen by a smoke run (Appendix D). Round five's recipe arm decays the learning rate to zero on a cosine schedule over the 20 epochs [12] and keeps the epoch with the highest validation directional recall among epochs meeting the stable floor.

Each candidate was converted as in Section 5.2 with two of the study's calibration selections, rebuilt on delta-encoded inputs from the study's pool (the 27,109 eligible Weinstock validation windows) and index lists: sequential-200 (diagnostic) and uniform-200, draw 101 (the candidate artifact). The validation split thus served both selection and INT8 calibration, and because the round-two winner and every later candidate trained on no Weinstock window, their INT8 calibration is cross-cohort. Confirmatory contrasts pair both models' predictions on the same windows, keyed by participant and anchor time, and use the paired participant-cluster bootstrap described above (2,000 draws, a seed frozen per round). Each contrast compares one candidate artifact, the best validation seed, with one champion artifact: the interval resamples participants, is conditional on both fitted models, and carries no training-seed variation.

7 Results: Calibration Coverage

7.1 Split and calibration provenance

Table 14 (Appendix C) summarizes the audit. No test window has a target in a training row, and none repeats a training window. The context-prefix overlap instead affects 17,266 test windows (13.6%) whose targets fall in rows that GlucoBench assigns to validation; validation labels were not used for fitting or checkpoint selection. Most validation frames begin with 192 training rows of model context, so 19,402 validation windows (35.1%) repeat training windows exactly.

The originally bundled artifact's calibration set is narrower than its size suggests: its 200 overlapping windows come from one participant and one segment, 211 distinct

Table 4: Measured coverage of the 200-window calibration sets (draw 101) and the resulting INT8 input range. The pool holds 27,109 eligible windows from 127 participants, range 39–401 mg/dL.

Strategy (200)	Part.	Readings	Max	INT8 range
Sequential	2	224	251	0–250.9
Uniform random	96	2,300	401	0–400.8
Balanced random	127	2,359	401	0–400.8
Range coverage	127	2,358	401	0–400.8

readings spanning 17.5 hours, and 178 of the 200 windows repeat training windows. Its converted input scale, 0.9408 mg/dL per code with zero point −128, gives an input range of 0–239.9 mg/dL, the quantized image of the largest calibration input, 240.0 mg/dL; Figure 3 (Appendix C) shows how much of the development data lies above that ceiling. These historical facts motivated the controlled study.

7.2 Coverage manipulation

The sampling interventions changed measured coverage, not just the strategy label (Table 4). The sequential-200 set draws from 2 of 127 pool participants and 224 unique readings with a maximum of 251 mg/dL (it differs from the historical first-200 set of Section 7.1 because the eligible pool excludes context-prefix and gap-filled windows); every randomized 200-window set covers 90–127 participants, roughly 2,300 unique readings, and the pool's full 39–401 mg/dL span. The INT8 input ranges split accordingly: 0–250.9 (scale 0.98 mg/dL per code) for sequential-200 against 0–400.8 (scale 1.57) for every randomized set, seed, and draw, a 60% coarser input grid as the price of the wider range. Sequential at budget 1,000 reaches 6 participants, 1,180 readings, and the full range.

7.3 Calibration effects on directional recall

Table 5 and Figure 2 give the main result on the held-out-participant component. Float LiteRT is prediction-identical to its PyTorch checkpoint at every seed. With the sequential-200 calibration, mean directional recall drops from 0.487 to 0.354 while accuracy *rises* from 0.507 to 0.570, the accuracy illusion the originally bundled artifact showed, reproduced under the corrected protocol; with any randomized 200-window selection, INT8 matches float to the third decimal (0.487). `falling_fast` is the most sensitive class: 0.550 float, 0.236 under sequential-200, 0.547 under uniform-200.

Table 6 reports the prespecified contrasts. The primary contrast, range coverage versus participant-balanced random at 200, is +0.06 percentage points (95% CI −0.02 to +0.13), but it could not test H1 (that broader input-range coverage beats random selection with the same participant allocation) as posed: both arms reach the pool's 39–401 mg/dL span and receive

Table 5: Held-out-participant metrics, means over five training seeds (and three draws for stochastic arms). Macro recall is the five-class mean recall (balanced accuracy); directional is the mean recall of the four non-stable classes.

Model	Accuracy	Macro recall	Directional
Float LiteRT	0.507	0.494	0.487
INT8, budget 200			
Sequential	0.570	0.412	0.354
Uniform random	0.496	0.492	0.487
Balanced random	0.497	0.491	0.487
Range coverage	0.497	0.492	0.487
INT8, budget 1,000			
Sequential	0.499	0.492	0.487
Uniform random	0.497	0.492	0.488
Balanced random	0.497	0.492	0.488
Range coverage	0.496	0.492	0.488

the same 0–400.8 mg/dL input range (Table 4), so they quantize every input identically and differ only in the windows that set the internal activation ranges. Uniform random versus sequential at 200, a prespecified secondary, is +13.38 points (CI +10.29 to +16.88). The effect appears at every training seed (13.05 to 14.14 points, standard deviation (SD) 0.47 across seeds), while the float model's own directional recall has an across-seed SD of 1.34 points and the three uniform draws within a seed an SD of 0.02 to 0.21 points. Within this shared-range regime, raising a randomized strategy's budget from 200 to 1,000 changes recall by at most +0.08 points (only the balanced-random interval, $[+0.02, +0.15]$, excludes zero), and the budget-by-strategy interactions among the three randomized strategies lie between −0.05 and +0.06 points (temporal: up to +0.07), every interval including zero. Sequential is the exception: its 1,000-window set reaches the full range and gains +13.35 points over its 200-window set, so its interaction with uniform random is −13.33. Given the same windows, the min–max observer moves seed-mean recall by at most 0.12 points (0.354 vs. 0.354 sequential; 0.488 vs. 0.487 range coverage). The temporal component mirrors every pattern (Table 6).

7.4 Mechanism: clipping, not resolution

Table 7 decomposes the sequential-200 loss with the input-grid controls. The clip-only image of the inputs, fed to the float model, reproduces essentially the entire 13.35-point float-to-INT8 difference (float−clip 13.136); re-gridding the clipped inputs moves it by only 0.01 points (float−QDQ 13.125), and the INT8 interior adds 0.23 points. Under the sequential-200 range, 27.8% of held-out windows have at least one clipped input slot by the rounded-code rule (60.9% of `falling_fast` windows, the class that collapses), against none under the randomized range: the 3.5% of held-out windows that contain the 401 mg/dL code exceed its real-valued top of 400.8 mg/dL, but that value rounds into the top code, so no input slot clips. Under uniform-200, the coarser

Table 6: Prespecified contrasts in mean directional recall (percentage points, 95% CI), paired participant-cluster bootstrap, 2,000 draws; no draw had an undefined endpoint. Interaction rows give the difference between two strategies' 1,000-versus-200 effects, a definition fixed after the primary analysis was published because `protocol.json` names the interaction without a formula. The study plan marks every secondary interval descriptive.

Contrast	Held-out	Temporal
Primary: range coverage vs. balanced random, 200	+0.06 [−0.02, +0.13]	+0.04 [−0.04, +0.13]
Range coverage vs. uniform random, 200	+0.01 [−0.05, +0.07]	+0.02 [−0.07, +0.10]
Uniform random vs. sequential, 200	+13.38 [+10.29, +16.88]	+12.43 [+10.28, +14.43]
Uniform random vs. sequential, 1,000	+0.05 [−0.05, +0.15]	+0.00 [−0.12, +0.12]
Uniform random: 1,000 vs. 200	+0.02 [−0.07, +0.10]	−0.03 [−0.12, +0.06]
Balanced random: 1,000 vs. 200	+0.08 [+0.02, +0.15]	+0.04 [−0.05, +0.12]
Range coverage: 1,000 vs. 200	+0.03 [−0.09, +0.16]	−0.01 [−0.11, +0.09]
Sequential: 1,000 vs. 200	+13.35 [+10.35, +16.77]	+12.40 [+10.23, +14.43]
Interaction: uniform random vs. sequential	−13.33 [−16.76, −10.29]	−12.43 [−14.46, −10.26]
Interaction: range coverage vs. balanced random	−0.05 [−0.16, +0.07]	−0.05 [−0.16, +0.06]
Interaction: range coverage vs. uniform random	+0.02 [−0.12, +0.16]	+0.02 [−0.11, +0.16]
Interaction: balanced random vs. uniform random	+0.06 [−0.04, +0.18]	+0.07 [−0.07, +0.20]

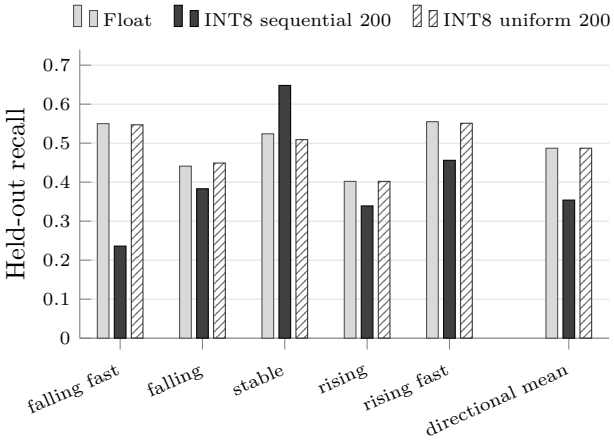


Figure 2: Per-class held-out recall, mean over five seeds (and three draws for the uniform arm). The sequential-200 calibration collapses `falling_fast` and inflates `stable`; a uniform random selection of the same size restores the float profile.

input grid alone changes float recall by +0.20 points (float−QDQ −0.20; no interval computed), so covering 39–401 mg/dL cost no measurable recall through resolution in this model.

7.5 Baselines

On held-out participants the five-seed float CNN reaches 0.487 directional recall against 0.278 for the recent-slope baseline and 0.461 for logistic regression on the same 12 readings (Table 15, Appendix C). The constant-`stable` classifier has the highest accuracy of all (0.727) and zero directional recall, which is why accuracy cannot rank these models.

7.6 Sensitivity tiers

Both headline findings hold in every frozen sensitivity tier and both components (Appendix C, Table 17): uniform random calibration beats sequential by +10.5 to +13.2 points of directional recall (widest single-tier intervals: held-out gap-filled inputs +11.44 [+6.56, +15.90]; temporal gap-filled endpoint +10.53 [+6.48, +14.56]), and the float-to-INT8 gap under uniform calibration is at most about 1 point (largest +1.04). Without windows whose anchor or label endpoint is a 39 or 401 mg/dL code, the held-out contrast stays at +13.27 [+10.25, +16.75] points.

Table 7: Mechanism controls on the held-out component (directional-recall differences in points, seed means). QDQ = float model on the artifact's quantize–dequantize input grid; clip = float model on inputs clipped to the artifact's range without re-gridding. Stochastic arms average all three draws; sequential has one order.

Calibration arm	Float−clip	Float−QDQ	QDQ−INT8
Sequential 200	+13.14	+13.12	+0.23
Uniform 200	+0.01	−0.20	+0.17
Range coverage 200	+0.01	−0.20	+0.16

7.7 Two further cohorts

Dubosson (nine participants with type-1 diabetes, readings to 400 mg/dL) and Hall (57 mostly non-diabetic participants, maximum 303 mg/dL) each received the reduced matrix described with the protocol. Each cohort's protocol froze one prediction, without a decision rule or margin: Dubosson's uniform−sequential difference would be positive; Hall's would be near zero, because with almost no readings above 240 mg/dL a sequential selection could not meaningfully under-cover the range. Dubosson's held-out component is a single participant, so its protocol designates the temporal component (six participants) and reports held-out descriptively; Hall designates held-out (six participants) and keeps temporal (50) secondary. Both designated contrasts' frozen percentile intervals include zero, and with six clusters cluster-based inference tends to over-reject

[53], so Table 8 replaces them with an exact sign-flip test over participants: each participant's additive contribution to the pooled difference (mean over seeds and draws; the contributions sum to the difference) has its sign flipped in all $2^6 = 64$ patterns, so the smallest attainable two-sided p is 0.031.

Each prediction's outcome is reported separately. For Dubosson, the designated difference is positive, +1.98 points, but one participant contributes +1.62 of it (the other five −0.04 to +0.16) and the sign-flip test gives $p = 0.094$: the point estimate has the predicted sign, and the data do not separate it from zero. For Hall, the designated held-out difference, +1.31 points ($p = 0.25$; one participant contributes +1.08, the other five −0.09 to +0.21), is inconclusive: without a frozen margin it supports neither a positive effect nor a near-zero one. The premise behind the Hall prediction failed: it reasoned from the share of readings above 240 mg/dL, the historical ceiling, but the sequential-200 set reached only 165.0 mg/dL. Hall's 50-participant temporal component shows +4.31 [+1.14, +7.34] points.

The following reading is exploratory. The mechanism predicts the effect from the share of windows a sequential ceiling clips beyond those uniform calibration clips, not from the distance in mg/dL between that ceiling and the cohort maximum. Weinstock's sequential-200 ceiling of 250.9 mg/dL clips 27.8% of held-out windows against none under uniform calibration, and costs 13.4 points. In the small cohorts the sequential selection's clipped share exceeds uniform calibration's by at most 4.3 percentage points (not at all for Dubosson's held-out participant), and the effects are 0.3 to 4.3 points. Among the three components with excess clipping the shares do not order the effects (Dubosson temporal has the largest excess, 4.3 percentage points, but a smaller effect than Hall temporal, excess 3.1), so no finer dose-response is claimed. Unlike Weinstock, uniform calibration itself under-covers both small cohorts: its ceilings are 352.7–354.5 mg/dL against Dubosson's 400 maximum and 189.9–233.9 mg/dL against Hall's 303, and it still clips 10.9% of Dubosson temporal windows. In every cohort the sequential selection again drew on two early participants, and uniform calibration sits within about a point of the float model. Per-seed records: `confirmation_dubosson.json`, `confirmation_hall.json`; contributions, sign-flip tests, and clipped shares: `panel_revision_r1.json`.

7.8 Delta-encoded inputs

If calibration failures are input-range failures, re-encoding the input should remove them. This post-hoc experiment on previously touched held-out data had gates frozen beforehand. Two variants were trained from scratch on the same eligible windows and seeds: 11 consecutive differences, and the same differences plus the last absolute value divided by 16, placing a coarse level channel inside the delta range. Both passed the frozen float gate (five-seed held-out directional recall ≥ 0.482): differences alone reach 0.497 and differences plus the scaled anchor 0.509, against the absolute-input model's 0.487. The frozen rule advanced the higher-scoring variant, so the anchor variant was chosen on held-out data (0.5088 against 0.4969), and its margin over the absolute model is selection-affected.

Because activations, the input included, are quantized per tensor, the anchor model's differences and level channel share one input step, set by the calibration windows: 0.247 per code under contiguous first-N selection (0.247 mg/dL for differences, 16× that for the level, which saturates at 320 mg/dL) and 0.231, 0.713 and 1.054 under the three uniform-200 draws (level ceilings 511, 833 and 776 mg/dL), against the absolute model's 1.572 mg/dL under uniform selection (`panel_revision_r1.json`). Under contiguous first-N selection, which costs the absolute-input model 13.4 points, the delta model loses half a point against its own float (+0.50 [+0.33, +0.67]), and uniform minus sequential is +0.54 [+0.38, +0.72] (Table 18, Appendix C). The frozen prediction of a near-zero difference again had no margin; the measured difference is a twenty-fifth of the absolute model's but excludes zero, so the encoding makes INT8 far less sensitive to calibration selection, not insensitive. The frozen ship criterion (sequential-INT8 within 0.5 points of float) passes at 0.4955 points, a margin of less than five thousandths of a point, while the interval above straddles the threshold. The delta pair then bundled used the uniform selection, under which the five-seed float and INT8 means agree to the third decimal (Table 18; bundling history in Appendix A). Full record: `delta_inputs_r1.json`.

7.9 Deployment costs

Calibration coverage has no deployment cost in this model. All 110 study INT8 artifacts serialize to exactly 11,976 bytes regardless of strategy, budget or observer (float: 17,352 bytes, a 31.0% reduction) and share a single operator inventory, so no change in operators or size is attributable to any calibration arm; the 40 second-cohort and 20 delta-encoded INT8 artifacts have the same size and operator inventory (`artifacts` in `panel_revision_r1.json`). The bundled 50,645-parameter pair serializes to 208,560 (float) and 61,936 bytes (INT8), a 70.3% reduction (`MODELS.md`). Offline preparation is also negligible: every sampler selects its windows in under 0.03 s, and INT8 export averages 0.5 s for a 200-window set and 0.8 s for a 1,000-window set on the study host.

Inside the artifact, calibration changes more than the input pair. Between the sequential-200 and uniform-200 artifacts of one checkpoint, 12 of 41 graph tensors differ: the graph input, the three activation tensors requantized between operators, the output, and the int32 bias of the final layer, whose scale follows its input's, plus six layout-operator copies of those parameters; the three

Table 8: Second-cohort results (directional recall; five-seed means, uniform over three draws), with the Weinstock held-out component for comparison. Seq. and Unif. are INT8 under sequential-200 and uniform-200; Δ is uniform−sequential in points; *marks the contrast each frozen protocol designates. The six-participant designated components get an exact two-sided sign-flip test (64 patterns) in place of their frozen percentile intervals, which are superseded: Dubosson temporal $[-0.00, +3.05]$ (1.2% undefined draws), Hall held-out $[-0.14, +3.30]$; no other component had undefined draws. Support counts `falling_fast/falling/rising/rising_fast` windows. Clipped: share of windows with an input above the INT8 range, sequential-200 / uniform-200 (mean of three draws), in all windows and in `falling_fast` (ff) windows.

Cohort	Component	Float	Seq.	Unif.	Δ	Inference	Support	Clipped, all	Clipped, ff
Weinstock	held-out (20)	0.487	0.354	0.487	+13.38	[+10.29, +16.88]	1,590/4,599/4,239/2,233	27.8 / 0.0%	60.9 / 0.0%
Dubosson	temporal (6)*	0.489	0.458	0.478	+1.98	$p = 0.094$	28/58/68/57	15.2 / 10.9%	75.0 / 42.9%
Dubosson	held-out (1)	0.481	0.472	0.475	+0.27	descriptive	19/106/115/15	0.0 / 0.0%	0.0 / 0.0%
Hall	held-out (6)*	0.393	0.375	0.388	+1.31	$p = 0.25$	41/261/303/50	3.4 / 0.5%	26.8 / 9.8%
Hall	temporal (50)	0.428	0.378	0.421	+4.31	[+1.14, +7.34]	34/463/411/118	4.1 / 1.0%	61.8 / 41.2%

int8 weight tensors are identical. The artifact is also less integer than its name: both convolutions, their bias additions, and the average pool run in float32 on dequantized tensors, and only the final fully connected layer runs in int8 (per-operator types in the same file). The prespecified iPhone benchmark (Table 19, Appendix C) matches this picture: the eight calibration arms differ by at most 0.05 μs in mean latency (3.42–3.47 μs), so the calibration fix carries no latency cost on this phone, and INT8 is slower than float in every arm (+0.64 to +0.70 μs), as expected of a graph that adds quantize and dequantize steps to the same float convolutions.

8 Results: Staged Accuracy Sweeps

Table 9 collects every confirmatory read; subsections report each round's selection and outcome. Differences are in points of the round's endpoint (recall or F1 ×100), and every quoted level names its artifact and split.

8.1 Round one: a macro-F1 winner rejected after its read

Round one asked whether loss design, capacity, window length, pooling across cohorts or seed ensembling improves the delta-encoded classifier (`protocol_sweep_r1.json`). Stage A compared four losses at the study architecture (validation macro-F1, three-seed means). A distance-penalized CE, which adds the expected class distance $\lambda \sum_j p_j\,|j - y|$ with $\lambda = 0.5$ to the weighted CE (a variant of class-distance-weighted CE [54]), scored 0.427, ahead of weighted CE (0.381), ordinal label smoothing (soft ordinal labels [55], here moving 0.15 of the target mass to the adjacent classes; 0.390) and focal loss with $\gamma = 2$ [56] (0.380). Stage B crossed channel widths with window length under that loss: (96,144) channels (42,725 parameters) reached 0.439, and a 2-hour window gained at most 0.04 points at any width. Pooling the Weinstock, Dubosson and Hall training windows (Stage C) lost 1.6 points, and a five-seed mean-logit ensemble (Stage D) did not beat the best single seed (0.4426 against 0.4430), so seed 1 of the (96,144) distance-penalized model became the candidate.

The read passed every gate (Table 9): the candidate's INT8 model beat the champion's by +5.68 macro-F1 points held-out and +5.52 temporally, within 0.1 point of its own float, and all five candidate seeds beat the champion (five-seed mean +5.25 points). The candidate was nevertheless not bundled: directional recall, the quantity the bundled models had been selected on, fell from 0.517 to 0.438 held-out and from 0.497 to 0.406 temporally. That drop is the basis of the rejection, a decision taken after the read (Table 13) that set the endpoint of every later round. A per-class read made after the gates were scored (Appendix D, Table 20) shows an operating-point shift toward `stable`: F1 rose in every class, half of the summed gain from `stable`, while recall fell in three of the four directional classes. Full record: `accuracy_sweep_r1.json`.

8.2 Round two: DiaData training under a directional endpoint

Round two returned to the weighted CE and asked whether more real data, and how it is mixed, raises directional recall (`protocol_sweep_r2.json`). Stage M compared five mixtures at (96,144) channels (three-seed validation means): Weinstock only 0.512; Weinstock plus the five DiaData training cohorts in natural proportions 0.527 or with equal windows per cohort 0.532; the five DiaData cohorts alone 0.535; and natural proportions with fast-class oversampling 0.526, level with natural proportions, as expected once class weights are recomputed on the oversampled labels. The DiaData-only mixture won with stable recall 0.540, above the floor; its model never sees a Weinstock training window. Narrower (24,36) and wider (192,288) channels lost (0.526 and 0.524), and seed 1 led the five at 0.539 (champion INT8: 0.519).

The read passed all three gates (Table 9): +0.60 points held-out, stable recall 0.459 against 0.475, and INT8 within 0.11 points of float, with the temporal component agreeing in sign. The interval includes zero, so gate 1 here certifies non-inferiority within 0.5 points, not superiority. The effect did not hinge on the cho-

Table 9: Confirmatory reads of the five sweep rounds. Each row compares one candidate artifact (uniform-200 INT8) with the champion's INT8 artifact on the same windows: the delta-encoded pair in rounds one and two, the round-two pair in rounds three to five (Table 12). Δ = candidate minus champion in points, paired participant-cluster bootstrap (2,000 draws), conditional on the two artifacts; for scale, round three's five candidate seeds spanned 1.1 points of float held-out directional recall. Cand. float and Cand. INT8 = the candidate's two models, whose gap the float-proximity gate bounds. Stable = INT8 stable recall, candidate/champion. Bold = pre-registered primary component; gates are scored there, and parentheses give what the same rules would give on a non-gating component. PEDAP rows are descriptive and carry no interval. Secondary macro-F1, candidate/champion INT8: 0.364/0.368 (round two, held-out), 0.341/0.364 (round three, held-out), 0.410/0.427 (round five, RT-CGM).

Round	Endpoint	Component	Cand. float	Cand. INT8	Champ. INT8	Δ [95% CI]	Stable	Gates
1	macro-F1	**held-out**	0.425	0.425	0.368	+5.68 [+5.16, +6.35]	—	pass
		temporal	0.417	0.418	0.362	+5.52 [+4.78, +6.15]	—	(pass)
2	directional	**held-out**	0.524	0.523	0.517	+0.60 [−0.06, +1.33]	0.459/0.475	pass
	recall	temporal	0.506	0.502	0.497	+0.50 [−0.37, +1.43]	0.520/0.537	(pass)
		PEDAP	—	0.541	0.522	—	0.394/0.403	descriptive
3	directional	**held-out**	0.531	0.531	0.523	+0.77 [+0.09, +1.49]	0.414/0.459	gate 2 fails
	recall	temporal	0.511	0.511	0.502	+0.97 [−0.01, +1.94]	0.481/0.520	(gate 2 fails)
		PEDAP	—	0.545	0.541	—	0.361/0.394	descriptive
4	no read: the candidate converts to SHA-256-identical copies of the champion's artifacts							
5	directional	**RT-CGM**	0.557	0.555	0.553	+0.20 [+0.11, +0.29]	0.605/0.607	pass
	recall	held-out	0.525	0.520	0.523	−0.32 [−0.75, +0.15]	0.436/0.459	(gates 1, 2 fail)
		temporal	0.509	0.505	0.502	+0.33 [−0.58, +1.24]	0.499/0.520	(gates 1, 2 fail)
		PEDAP	—	0.539	0.541	—	0.391/0.394	descriptive

sen seed: the five candidate seeds averaged +0.55 points above the champion artifact on held-out, none below it. Macro-F1 slipped held-out, and on PEDAP (7.0M windows, on which no model trained) directional recall rose. The candidate was bundled after golden parity passed in the Android unit tests on the Java virtual machine (JVM), on the iOS simulator and on the iPhone 18 Pro Max (51 tests, INT8 bit-exact). Full record: `accuracy_sweep_r2.json`.

8.3 Round three: RT-CGM, time of day and a temporal convolutional network

Round three staged three changes against the round-two recipe, five seeds per arm (`protocol_sweep_r3.json`). The data stage reopened RT-CGM, DiaData's copy of the JDRF CGM randomized trial [19], whose primary report randomized 322 intensively treated children and adults with T1D in three age groups (8–14, 15–24 and ≥25 years); the cohort name denotes this trial, not current real-time CGM. Appendix B gives its sensors, the cadence re-audit that admitted it, and the 355 of its 451 subjects with scored windows.

Adding RT-CGM to training lost 0.6 points of validation directional recall (five-seed means), and sine and cosine time-of-day input channels lost 0.3 (timestamps taken as device-local time). The architecture stage replaced the CNN with a causal dilated temporal convolutional network (TCN) [43]: four residual blocks, kernel 3, dilations 1–8, 72 channels and 63,005 parameters. It edged the CNN (0.533 vs. 0.532), and seed 2 led at 0.537.

The TCN's INT8 model beat the champion's by +0.77 points held-out, the first fully positive interval on the directional endpoint, agreed temporally (+0.97) and sat 0.05 points above its float, but failed gate 2: stable recall 0.414 against 0.459 held-out (−4.4 points against gate 2's 2-point limit; −3.8 temporally). Macro-F1 also fell, and on PEDAP directional recall rose slightly while stable recall fell (Table 9). The candidate was rejected and the round-two pair stayed bundled. Two selection details shaped round five: seed 2 was picked by strict argmax although seed 1 trailed by 0.01 points of directional recall but had higher stable recall (0.536 against seed 2's 0.526), and the 5-point selection floor admitted a seed that a 2-point floor would already have excluded on validation. The +0.77 points also belonged to the chosen seed: across the five candidate seeds the mean held-out effect was +0.05 points, three of five seeds were ahead, and their float held-out directional recall spanned 1.1 points. Full record: `accuracy_sweep_r3.json`.

8.4 Round four: a label-definition stability check and an identity outcome

Round four took its arms from the champion's error structure, read descriptively from round three's saved held-out predictions (Table 13): 72% of champion errors were adjacent-class, and slow-class errors went to the same-direction fast class (47% of `falling` errors, 46% of `rising`) more often than to `stable` (25% each). That read chose arms only; selection stayed on validation.

Before any arm trained, the round compared two labels on the 27,109 validation windows (`exp0_sweep_r4.json`). The endpoint label classifies $(g_{t+15} - g_t)/15$; an alternative label classifies the least-squares slope over the four readings $g_t, \ldots, g_{t+15}$; both use the trend-arrow rate

bands (±1 and ±2 mg/dL/min). The slope label recovers the endpoint label's directional classes with mean recall 0.954 (`stable` 0.995), and recomputing the endpoint label after adding independent Gaussian noise of $\sigma = 1$ or 2 mg/dL to the readings gives 0.915 and 0.860. This is a label-definition stability check. Both labels are functions of the same future readings, so their agreement bounds nothing that a model seeing only the 12 past readings can reach, and the recall attainable from CGM-only inputs remains unknown. White noise of this size is also gentler than published CGM error models, which add blood-to-interstitial lag, sensor-calibration drift and autoregressive noise [57], and the labels are defined on the sensor trace, which itself lags blood glucose. The abort rule, committed in the round's design specification before training, would have stopped the round had the slope label's agreement exceeded the champion's INT8 validation directional recall (0.533) by less than 2 points; with both labels sharing the future readings it could hardly fire, and it did not.

Three stages then varied the objective, the training targets and inputs, and weight averaging against the round-two recipe at five seeds per arm (Appendix D, Table 21); every arm lost to the weighted-CE control on validation directional recall. Square-root class weights lowered it to 0.405 while stable recall rose to 0.827, so the full inverse-frequency weights, with the slow-to-fast over-prediction they cause, are what buys directional recall here. A Huber regression head on the rate, decoded at the same rate bands, collapsed toward `stable` (0.200 directional, 0.950 stable), the pull toward the dense region of the target that regression on imbalanced targets is known to suffer [58]. A head for conditional ordinal regression (CORN) [59] reached 0.470. Slope labels for training changed little (−0.2 points), and $\sigma = 2$ mg/dL input noise traded directional for stable recall (−3.7 points). A uniform average of the five seeds' weights collapsed (0.238 directional, 0.003 stable), the expected outcome: weight soups average models fine-tuned from one pre-trained start [60], and independently initialized runs need not share a linearly connected basin [61].

The surviving candidate, the control arm's best seed, is the champion's own checkpoint and converts to SHA-256-identical artifacts (`96eef43c…/a887ea4e…`). A contrast would have compared the model with itself, so no confirmatory read was spent and, beyond the arm-choice read above, no held-out, temporal or PEDAP label was read; the round-two pair stayed bundled. Full record: `accuracy_sweep_r4.json`.

8.5 Round five: the seed-4 flatten-head artifact

Round five tested how the final convolutional features reach the classifier. The global average pool can erase where in the hour a movement happened: two synthetic histories carrying the same 10 mg/dL rise, placed 30 versus 20 minutes before the last reading, produce bit-identical outputs from the champion's checkpoint and from both of its bundled artifacts (`recency_collision.json`, Section 11), an existence proof rather than a rate on real data. A recipe stage compared the round-two recipe with the cosine schedule and validation-chosen kept epoch of Section 6.2; a head stage then replaced the pool, under the winning recipe, with a flatten head in which the 144×12 feature map feeds a 1,728-to-5 linear layer (50,645 parameters against 42,725); a final stage picked the seed (`protocol_sweep_r5.json`). Selection took the cosine recipe (five-seed validation means 0.535 against 0.532), then the flatten head (0.538 against 0.535), and seed 4 led at 0.541 directional and 0.546 stable recall, the only flatten-head seed above the champion checkpoint's own validation float (0.539). These numbers are selection-biased: the kept epoch and the seed were chosen on the split they are quoted from. The candidate differs from the champion in head, learning-rate schedule, kept epoch and seed at once, and its INT8 artifact is calibrated with uniform-200 from the Weinstock validation pool, a cohort it never trained on.

The read moved to RT-CGM, whose labels no round had evaluated: 12.67M eligible windows from the 355 of 451 admitted subjects with scored windows. Round three had trained a rejected arm on this cohort, but neither model under comparison trained on it, so the comparison is zero-shot for both. All three gates passed (Table 9): the candidate's INT8 model beat the champion's by +0.20 directional-recall points [+0.11, +0.29], a fifth of the 1.1-point float spread of round three's five seeds on Weinstock held-out, held stable recall within 0.2 points (0.605 against 0.607) and sat 0.25 points under its own float. That gap includes any cost of input clipping: both artifacts share one INT8 input range, which 2.8% of RT-CGM windows exceed (the level channel saturates at 511 mg/dL). RT-CGM readings span 14–654 mg/dL (Table 11); DiaData does not record the source of values outside the 40–400 mg/dL span the Dexcom cohorts' codes bound, and without the 1.2% of windows holding one the gain is +0.29 points [+0.21, +0.38] (`panel_revision_r1.json`). The candidate replaced the round-two pair under the rule fixed before the read, but the result is a choice between two artifacts, not evidence for a design: round three's chosen-seed effect (+0.77) shrank to +0.05 at the five-seed mean, and the five-seed RT-CGM comparison of the two recipes, which would measure seed variation here, was not run. The four changed factors are not separately identified. The non-gating secondaries carry costs: on both Weinstock components the candidate fails gates 1 and 2 (held-out −0.32 points with stable recall 0.436 against 0.459; temporal +0.33 with 0.499 against 0.520), macro-F1 is lower on all three paired splits, including RT-CGM (0.410 against 0.427), and PEDAP directional recall is 0.539 against 0.541. PEDAP's per-class changes show where the predictions moved: `falling_fast` recall rose 6.7 points and `rising_fast` 2.3, while `falling` fell 6.5,

`rising` 3.1 and `stable` 0.3; the candidate labels 16.0% of PEDAP windows `falling_fast` against the champion's 12.3%, and its `falling_fast` precision drops from 0.284 to 0.242, consistent with an operating-point shift from the slow to the fast classes rather than better discrimination. Full record: `accuracy_sweep_r5.json`; PEDAP per-class detail: `panel_revision_r1.json`.

9 Discussion

9.1 Input-range coverage, not sample count

That calibration data matter in PTQ is known [37, 39–41]; the converter guidance quoted in Section 2.4 asks for samples "randomly chosen, in no particular order" [10], and the sequential arm contravenes it by taking contiguous windows in stored order. What this study adds is a measurement of that departure where it is easy to make: stride-one CGM windows overlap, so 200 contiguous windows sit inside about 17 hours of one or two sensors' readings and their input range stops below the population's tail. At identical size, the sequential and uniform sets differ by 13.4 points of held-out directional recall, concentrated in `falling_fast` and placed at the input edge by the clip-only control, while raw accuracy rises. Sample count did not predict the loss: the 1,000-window sequential set recovers fully with 6 participants because it reaches the full range.

The rule this evidence supports is to cover the deployment input range, which any randomized selection does in this pool: every randomized 200-window set reaches 39–401 mg/dL and receives the same 0–400.8 mg/dL input range. That shared range also limits what the null contrasts show. The primary contrast and the randomized budget contrasts compared arms whose inputs were quantized identically, so they show that, once the range is covered, stratification, a larger budget, and the observer choice moved recall by at most about a tenth of a point, not that range-aware selection is useless. It was not tested where uniform draws under-cover the range, as in Dubosson and Hall, whose uniform-200 sets still clip 10.9% (Dubosson temporal) and 0.5–1.0% (Hall) of windows (Table 8). A more direct remedy would remove the input's dependence on calibration data by fixing its range from the sensor's reporting limits (here 39–401 mg/dL, codes included). The PT2E API offers fixed quantization parameters (torchao's `FixedQParamsQuantizationSpec`), but LiteRT Torch applies them only to sigmoid and softmax outputs, so the input would need a custom annotation, a route untested here. Unlike difference encoding, which left a +0.54-point dependence (Section 7.8), fixed parameters would remove it.

The absent resolution cost is what 8-bit error analysis predicts [33, 37]. Spreading 0–400.8 mg/dL over 255 steps gives a step of 1.572 mg/dL, a maximum rounding error of 0.79 mg/dL, and a root-mean-square error of about 0.45 mg/dL (the step over $\sqrt{12}$), small against the 15 mg/dL change over 15 minutes that separates `stable` from `falling` or `rising`; clipping at 250.9 mg/dL removes up to 150 mg/dL from a reading.

Two accounts of the class asymmetry fit the mechanism controls (Section 7.4), which localize the sequential-arm error at the input edge, and the study does not separate them. In one, fast falls start high — 61% of `falling_fast` windows contain a reading above the sequential range — and a model that cannot see how high a trajectory started loses a cue that a steep fall is coming. In the other, clipping maps every reading above 250.9 mg/dL to one code, flattening the clipped part of a high fast fall into a plateau and erasing its slope as well as its level; this account would also explain the inflated `stable` predictions and fits a label defined by the difference between readings.

9.2 Scope of the accuracy illusion

The sequential-arm INT8 model is a worse directional classifier with better accuracy (0.570 vs. 0.507): clipping pushes predictions toward `stable`, the class holding three quarters of the weight, so any pipeline that gates a quantized model on raw accuracy would have promoted the broken artifact. Directional (minority-class) recall or five-class macro recall exposes it immediately (0.354 and 0.412 against float 0.487 and 0.494, Table 5) [7]; this is the practical reading of the study for small imbalanced time-series deployments. Not every class-balanced metric guards against a shift toward `stable`: round one's macro-F1 rose under one (Section 8.1).

Parity and runtime agreement are necessary but orthogonal checks: the three clients agree bit-exactly on dequantized INT8 outputs for the tested vectors, yet parity says nothing about whether the artifact's input range fits the population, and validates neither the labels, the cohort split, probability calibration, clinical safety, nor usefulness to a person with diabetes.

9.3 What the sweeps show

The sweep program's main lesson is that a pre-registered endpoint can pass every gate and still optimize the wrong thing. Round one's F1 rose in every class while directional recall, the quantity the bundled models had been selected on, fell by 7.9 points; pre-registration fixed how the endpoint was analyzed, not whether it was the right endpoint. From round two on, the gates encode that quantity, directional recall under a stable-recall floor, but the floor is relative to each round's champion. Held-out INT8 stable recall therefore drifted from 0.475 to 0.459 to 0.436 across the two promotions, the second gated on another cohort; a floor anchored to a fixed reference would prevent that drift.

What the rounds established is narrower than five rounds of selection suggest. Round one's gain held at every seed and lay on the wrong axis. Round two's

DiaData-only recipe matched or beat the delta-encoded champion artifact at every seed on held-out, the most consistent result of the program, though on a reused split and with a confirmatory interval that includes zero. Round three showed how much seed choice alone can contribute (+0.77 points for the chosen seed, +0.05 at the five-seed mean), and round four's arms all lost on validation, two in ways prior work predicts. Round five showed that one artifact passed the pre-registered gates on a cohort no round had evaluated, by +0.20 points against the 1.1-point float spread of round three's five seeds on Weinstock held-out, while failing gates 1 and 2 on both Weinstock components and lowering macro-F1 on RT-CGM (0.410 against 0.427), with PEDAP a near-tie (0.539 against 0.541). It did not show that a flatten head, a cosine schedule or validation-chosen epochs improve this task: the four factors changed together, one seed per side was compared, and the five-seed RT-CGM comparison was not run. No round measured the recall attainable from 55 minutes of CGM history.

A displayed class is the model's estimate of the rate band that the sensor-reported glucose will average over the next 15 minutes, neither a measurement of the current rate nor a reading of blood glucose. On both evaluated cohorts (Appendix D, Table 22), 63% of displayed movement classes were followed by a mean rate within ±1 mg/dL/min, while a displayed `stable` was right in 92% (held-out) and 94% (RT-CGM) of windows; a displayed `falling_fast` was followed by a falling or fast-falling rate in a third (held-out) to a half (RT-CGM) of windows; and 9–15% of truly falling or rising windows were shown in the opposite direction. Every movement class is predicted 1.7 to 5.2 times as often as it occurs: `falling_fast`, the most over-called, pairs the highest recall with the lowest precision (0.706 at 0.136 held-out, 0.673 at 0.214 on RT-CGM), while `rising_fast` has the lowest RT-CGM recall of any class (0.483); inverse-frequency class weights and a directional-recall endpoint select this operating point (Table 21). Below 70 mg/dL the model almost never recognized a fast fall (0 of 1 held-out and 2 of 96 RT-CGM `falling_fast` windows), which alone makes that band's directional recall the lowest (0.33 held-out, 0.36 RT-CGM); over the other three directional classes the ≥250 mg/dL band is lowest (0.38, 0.44). These figures describe a research classifier on retrospective data; they are not evidence about how a person would use its output.

10 Limitations

1. This is a locked, corrected reanalysis, not a fresh test. The held-out participants were scored once before under the flawed aggregate protocol, and the five-class decision was informed by a historical test run. The protocol, samplers, and endpoint were frozen before the new held-out labels were read, but prior exposure cannot be undone by re-splitting [21].
2. The calibration findings rest on one architecture, one quantizer, and one primary cohort: a case-control sample of older adults with long-standing type-1 diabetes, about half of whom had a severe hypoglycemic event in the prior year, recorded with blinded CGM [17]. The range-stratification and budget nulls hold in a saturated regime in which every randomized 200-window selection receives the same INT8 input range; range-aware selection was not tested where random draws under-cover the range. These nulls and the observer comparison describe this 2,909-parameter CNN with per-tensor affine input quantization, not PTQ in general [22, 35].
3. The second-cohort evidence is exploratory: neither designated six-participant contrast is significant at the 5% level under an exact sign-flip test, and Hall's temporal component is positive against a frozen near-zero prediction whose premise failed (Table 8). All other calibration contrasts are single-cohort [4].
4. Stride-one windows remain overlapping and dependent [20], so intervals come from participant-cluster bootstraps conditional on the fitted models and calibration sets (Section 6.1), and six-participant components get an exact sign-flip test instead of a valid interval.
5. The accuracy sweeps reuse the main study's Weinstock splits: every round selected on its validation windows; rounds one to three read its held-out and temporal labels; round four's arm choice, an external review's offset scan and this revision's reanalyses reread them descriptively; and round five reports them again as non-gating secondaries (Table 13). Round five is an artifact-level ship decision between two artifacts (Section 8.5): the winner's validation numbers are selection-biased, its macro-F1 is lower on all three paired splits, and the five-seed RT-CGM comparison that could credit the flatten head or the cosine schedule was not run. No round had evaluated RT-CGM labels before, though a rejected round-three arm trained on it.
6. The gap-fill mask removes windows around missing readings, but the retained grid still contains formatter re-gridding between adjacent readings, and interpolation precedes windowing, so retrospective preprocessing may use information unavailable in real time. The frozen sensitivity tiers were evaluated after the primary analysis was published, and the gap-filled-endpoint tier's labels score agreement with the formatter's interpolated grid, not with unobserved sensor readings. Readings of 39 and 401 mg/dL, consistent with LOW and HIGH codes, are treated as glucose values, so a window that starts or ends at one carries only a bound on its rate (Table 17 excludes such windows in a check).
7. The five labels are deterministic transformations of the sensor trace, not expert annotations, and

that trace lags blood glucose through blood-to-interstitial diffusion and carries sensor-calibration drift and autocorrelated noise [57]. Round four checked only label-definition stability, also under independent Gaussian noise of $\sigma = 1$ and 2 mg/dL, a gentler perturbation than published autoregressive sensor-error models [57]; the recall attainable from 12 past CGM readings is unknown. The CGM-only model omits meal, insulin, activity, illness, and other covariates, and probability calibration and event-level performance were not evaluated.

8. Wear and therapy regimes differ across cohorts. Weinstock participants wore blinded CGM; most DiaData cohorts mix masked run-in and real-time wear, which DiaData does not separate, and DCLP3 and PEDAP include closed-loop insulin delivery (Table 11). The future 15-minute trajectory is therefore not the same physiological object across cohorts, and a deployed trend display would itself change the wearer's behaviour and so the trajectories it predicts. PEDAP randomized preschool children to automated insulin delivery or standard care [18]. Every cohort behind the bundled model has type-1 diabetes; no type-2 diabetes, pregnancy, or inpatient data entered its training or evaluation.
9. Device evidence is narrow. Of the 110 study INT8 artifacts, 8 (with one float artifact) were timed on one iPhone; the other 102, and the 60 INT8 artifacts of the second-cohort and delta-encoding runs, were not device-timed, though all 170 share one byte size and one operator inventory. Repeated calls on one phone are not evidence across devices, the historical Android device is no longer available, initialization was measured within-process, and energy and memory were not measured (Appendix C). INT8 showed no mean-latency advantage over float in any measurement; its size saving is 31% (5,376 bytes) for the 2,909-parameter study model and 70% (146,624 bytes) for the 50,645-parameter bundled model. The quantization trade-off studied here is an accuracy risk against a size saving, with no measured speed benefit.
10. The application replays synthetic or locally extracted historical traces. It does not ingest a live sensor stream and has not undergone prospective, external, usability, or clinical validation.

11 Reproducibility and Ethics

11.1 Code, artifacts, and computational record

The repository contains the training, conversion, benchmark, Android, iOS, and test code. The bundled artifacts are the round-five seed-4 flatten-head pair (cosine schedule, validation-chosen epoch), whose confirmatory protocol was frozen at repository commit `f9010b6` before the read. Their SHA-256 digests are `bfb37468...6153` (float) and `eacc0a74...f554` (INT8); full hashes are stored beside the assets and asserted by the golden-parity tests, and the application's model manifest (`MODELS.md`) records every bundled pair's hashes and provenance (Appendix A lists the pairs in turn). Python unit tests cover label boundaries, segment-aware windows, model shape, conversion helpers, synthetic replay behavior, and the provenance audit's split bookkeeping. Android unit tests cover replay, buffering, quantization, latency aggregation, and view-model behavior; the iOS suite covers the same units plus golden-vector parity and runs in continuous integration on a simulator.

The conversion requirements record LiteRT Torch 0.9.1, TorchAO 0.17.0, and Python LiteRT 2.1.5, and document PyTorch 2.12.1 and NumPy 2.5.0 in the historical environment (the top-level requirements pinned older versions); these records are not a complete environment lock. PyTorch seed 0 and the formatter's split seed 0 control separate randomization steps; neither guarantees bitwise retraining across platforms. The batch sizes of Section 6.2 are recorded as `sweep_settings` in `panel_revision_r1.json`. The provenance audit's aggregate output, including the GlucoBench commit and a SHA-256 hash of the Weinstock data file, is versioned as `paper/results/split_audit.json`.

The calibration-coverage study is versioned under `paper/results/calibration-coverage/`: the frozen protocol, the sanitized environment (`environment.txt`; conversion pins, PyTorch 2.12.1, NumPy 2.5.3, reconciling the previously recorded drift), the eligibility audit, the complete aggregate results with all 100 planned cells (`aggregate_r1.json`), the artifact summary, and the SHA-256 hashes of all 190 study, second-cohort, and delta-encoding artifacts (`artifact_hashes_r1.json`). Re-analyses of cached results are in `panel_revision_r1.json` (`experiments/panel_revision.py`). Checkpoints, window keys, per-window predictions, per-participant confusion counts, and run reports stay local because they derive from participant-level data distributed under the providers' terms, so a third party must regenerate them with the committed runners before recomputing any interval. The committed analysis commands then recompute the reported numbers from those predictions, but because retraining is not bitwise reproducible across platforms, a third party's numbers may differ from the published ones. Accuracy-sweep rounds one, two, three, and five each version a frozen protocol and a full record in the same directory (`protocol_sweep_r`N`.json`, `accuracy_sweep_r`N`.json`). Round four, which spent no read, versions its record (`accuracy_sweep_r4.json`) and its label-definition stability check (`exp0_sweep_r4.json`); its design and pre-registered abort rule are in the round-four specification (commit `c8f3de6`). The synthetic diagnostic that motivated round five, `recency_collision.json`, is versioned with its reproduction

script in `paper/analysis/accuracy-review-2026-10-13/`.

11.2 Data availability and research-use scope

The CGM data are public research data. The Weinstock [17], Dubosson [51], and Hall [52] cohorts are accessed through GlucoBench under the source datasets' reuse terms [1]. The sweep cohorts come from DiaData v3 [11] under CC BY-NC 4.0, using only its raw (non-imputed) five-minute archive: CITY [62], DCLP3 (DiaData label DLCP3) [63], REPLACE-BG [64], SENCE [65], and WISDM [66] for training; PEDAP [18] as a zero-shot probe; and, as round five's primary endpoint, RT-CGM, the JDRF CGM randomized trial [19], whose DexCom SEVEN, MiniMed Paradigm and FreeStyle Navigator sensors (`diadata_audit_r1.json`) cannot be assigned to subjects (Table 11). Some DiaData cohorts originate from the Barbara Davis Center, Jaeb Center for Health Research, Joslin Diabetes Center, T1D Exchange, University of Colorado, and University of Virginia; the analyses, content, and conclusions presented here are solely the responsibility of the author and have not been reviewed or approved by these institutions. No CGM data are committed to the GlucoEdge repository or the distributed applications, and the committed replay trace is synthetic. The bundled weights were trained on DiaData cohorts released under CC BY-NC 4.0, and their INT8 quantization parameters were calibrated on Weinstock validation windows; the weights are released for non-commercial research use only. This work makes no diagnosis, monitoring, treatment, or dosing claim. Its outputs must not be used for medical decisions. The application displays this restriction in the primary interface.

The repository's `MODEL_CARD.md`, structured after Mitchell et al. [67], states the bundled model's intended use (offline research replay), out-of-scope uses (live decision support, alerts, dosing, and any clinical use), input requirements, evaluation results, and known failure modes, among them frequent movement labels during stable glucose and over-calling of the fast classes, together with the populations its data do not represent.

Code, bundled model files, and synthetic golden vectors are in the project repository; source CGM access is described by GlucoBench [1] and DiaData [11]. This secondary analysis used only public, de-identified data under the providers' terms; no participants were recruited, and no ethics review was sought.

12 Conclusion

On a corrected, provenance-masked protocol with five training seeds and 100 paired INT8 artifacts, the input range covered by the quantization-calibration windows decided whether static INT8 quantization damaged this compact CGM trend classifier. Contiguous first-N selection, whose 200 windows span two participants and cap the input range at 250.9 mg/dL, cost 13.4 points of held-out directional recall in a prespecified secondary contrast while raising raw accuracy — an illusion that directional or macro recall exposes. Any randomized selection of 200 windows restored float-level directional recall, and clip-only controls attributed the loss to input-range clipping, not grid resolution. What mattered was covering the deployment input range, which any randomized selection did in this pool; once it was covered, stratification, budget, and observer choice moved recall by at most about a tenth of a point. These nulls, the null primary contrast among them, are scoped to that saturated regime: range-aware selection was not tested where random draws under-cover the range, as they did in the two small cohorts. The corrected calibration is free at deployment; INT8 brought no mean-latency advantage in any measurement and saves 31% of file size for the 2,909-parameter study model and 70% for the bundled model.

The staged sweeps add a methodological lesson. Four rounds spent one pre-registered confirmatory read each; round four, whose candidate converted to the champion's own artifacts, spent none. Gates alone were not enough: round one's macro-F1 winner passed every gate while held-out INT8 directional recall fell from 0.517 to 0.438, and its post-hoc rejection made a directional endpoint and a stable-recall floor part of every later protocol. Under those rules, DiaData-only training passed round two's gates, with an interval that includes zero, and was bundled; no round-three or round-four arm was adopted. Round four's label-definition stability check compares two labels computed from the same future readings, so it bounds nothing that a model of past readings can reach; the recall attainable from CGM history is unknown. Round five was a choice between two artifacts: the seed-4 flatten-head artifact (cosine schedule, validation-chosen epoch), now bundled in the research applications, passed all gates on never-evaluated RT-CGM by +0.20 points, a fifth of round three's 1.1-point float seed spread on Weinstock held-out (its own RT-CGM seed spread was not measured), but failed gates 1 and 2 on both Weinstock components and lowered RT-CGM macro-F1 (0.410 against 0.427); pediatric PEDAP directional recall was a near-tie (0.539 against 0.541). Its head, schedule, kept epoch, and seed changed together and are not separately identified.

The calibration conclusions cover one primary cohort, architecture and quantizer, on a reanalysis of previously touched data; both headline findings hold in every frozen sensitivity tier. On two small further cohorts the frozen predictions were not confirmed: Dubosson's designated difference was +2.0 points (exact sign-flip $p = 0.094$), and Hall's +1.3 points ($p = 0.25$) was inconclusive, its premise failing when the contiguous selection reached only 165 mg/dL. That evidence is exploratory. GlucoEdge is a research prototype, not a medical device.

Declarations

This work received no external funding, and the author declares no competing interests. Author contributions (CRediT): the sole author, Mohamed Menasy, takes responsibility for all contributor roles. The code is released under the MIT License and the bundled model weights under CC BY-NC 4.0.

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A Provenance and Revision History

A.1 Bundled artifacts over time

The calibration study starts from the originally bundled application checkpoint (seed 0, historical protocol), whose INT8 artifact the project's conversion script calibrated on the first 200 validation windows in dataset order (Section 7.1). After the study, the applications' bundled models were replaced by its seed-0 float artifact and uniform-200 INT8 artifact; then by the delta-encoded pair after that experiment passed its frozen gates (Table 18), with both clients applying the delta transform before quantization; then by the round-two sweep winner; and finally by the round-five winner. Each bundled pair passed both golden-parity suites bit-exactly on the continuous-integration simulators and on-device golden parity on the iPhone (INT8 bit-exact; Appendix C); hashes and provenance are recorded in the application's model manifest at each step.

A.2 Earlier revisions

After the July results, neither the September revision, the provenance audit that followed it, nor the 10 October iOS revision retrained a model, re-evaluated one, or touched held-out glucose values. September reused the July results and computed its descriptive summaries from the rounded tables; the audit reran only the GlucoBench formatter and window construction; and the iOS revision added the client and its device observations on unchanged artifacts with the same SHA-256 hashes.

Those revisions reported accuracy 0.5184 and macro recall 0.5060 for float and 0.5866/0.4135 for INT8. These numbers came from a different denominator — the uncorrected aggregate frame, with context-target overlap and no gap mask — and from the original sequential calibration, so they are not comparable to Table 5 row by row. The qualitative pattern they showed (accuracy up, every directional recall down) is what the corrected protocol reproduces for the sequential arm.

A.3 Historical device observations and parity runs

Before the study, the originally bundled artifacts were observed on two phones (Table 10). On a Samsung Galaxy S22 Ultra (SM-S908E, Android 16), latency was measured inside the application's `classify` method with `System.nanoTime`: buffer writes, input quantization when needed, model execution, and output reading/dequantization, stopping before softmax, class selection, and construction of the returned prediction (model loading and user-interface updates excluded). The float statistics use the latest 100 inferences, the recorded INT8 statistics 21 after a model switch, and p95 is the sorted sample at zero-based index $\lfloor 0.95(n-1) \rfloor$, so the estimates differ in rank resolution. These timings measure neither the full classification call nor the kernel alone, and the runs were not controlled for a benchmark comparison. The iOS client timed the same region with the monotonic `DispatchTime` clock on the iPhone 18 Pro Max during 16× replay of the synthetic trace (one inference every 312.5 ms), the latest 100 inferences per model in two rounds, in a debug application build over the prebuilt release LiteRT binary; a separate test-runner benchmark called each model 100 times back to back after one warm-up call, in three runs. Neither phone showed an INT8 latency advantage (iPhone back-to-back means 2.83–2.85 μs float, 3.69–3.72 μs INT8), the same picture as the study's benchmark.

Table 10: Historical artifact and physical-device observations for the originally bundled models (uncorrected protocol era). Timing ends before softmax and class selection. Android sample counts differ by model; iPhone ranges span two rounds of 100 inferences per model. KiB = 1,024 bytes.

Measurement	Float	INT8
Artifact size (bytes)	17,352	11,976
Artifact size (KiB)	16.95	11.70
Galaxy S22 Ultra, Android 16, LiteRT 2.1.0		
Mean latency (ms)	0.334 ($n = 100$)	0.484 ($n = 21$)
p95 latency (ms)	0.540	0.665
Golden vectors passed	20/20	20/20
iPhone 18 Pro Max, iOS 27.0.1, LiteRT 2.3.0		
Mean latency (ms)	0.062–0.073	0.101–0.110
p95 latency (ms)	0.103–0.134	0.178–0.182
Golden vectors passed	20/20	20/20

All three Android instrumentation tests passed on the physical device: the `CompiledModel` path loaded both artifact hashes, matched float logits within 10^{-5} on all 20 golden windows, and matched INT8 dequantized outputs exactly. These July device tests ran an earlier Kotlin version that rounded ties toward positive infinity; the golden-vector inputs, given to 0.1 mg/dL, contain no tie at the original input scale, so the expected outputs are unchanged. All four iOS parity tests passed on the iPhone with the same result — artifact hashes, class-name order, float logits within 10^{-5}, and bit-exact INT8 outputs — using the half-to-even input rounding the July Android run predates. Python, Kotlin, and Swift therefore agreed on the tested vectors; model outputs may still differ on untested inputs, although input quantization now follows the same rounding rule on both sides.

B Cohorts, Model Variants, and Held-Out Read Ledger

Table 11 characterizes every cohort used in this work, Table 12 the model variants bundled at some point, and Table 13 every analysis that read held-out, temporal or RT-CGM labels, in date order.

RT-CGM needs more provenance than the table holds. The DiaData audit lists DexCom SEVEN, MiniMed Paradigm and FreeStyle Navigator sensors (`diadata_`

Table 11: Cohorts used in this work. Study = calibration-coverage study; RN = sweep round N. n = subjects in the data used (DiaData counts differ from the trials' randomized counts by up to three, except RT-CGM: 451 against the primary report's 322). Years = enrolment to end of follow-up in the primary report. Wear: blinded = readings hidden from the wearer; mixed = blinded or masked periods plus real-time wear, which the data cannot separate. Range: 39 and 401 mg/dL are consistent with LOW/HIGH codes of receivers that report 40–400 mg/dL; RT-CGM gives the range of window anchors. — = not verified from a primary source.

Cohort	Role	Source	n	Age (y)	Diabetes; therapy	CGM; wear	Years	Range (mg/dL)
Weinstock (GlucoBench)	Study: training, calibration pool, held-out and temporal evaluation. R1: training. All rounds: selection and INT8 calibration pool. Held-out and temporal reads: primary in R1–R3, secondary in R5	[17]	200 (192 after formatting)	≥60	T1D for ≥20 y; case-control on severe hypoglycemia; —	Dexcom G4 per GlucoBench [1]; blinded	—	39–401
Dubosson (GlucoBench)	Study: second cohort. R1: pooled-training arm	[51]	9	—	T1D; —	Medtronic iPro2 per GlucoBench; professional (blinded)	—	to 400 (from mmol/L)
Hall (GlucoBench)	Study: second cohort. R1: pooled-training arm	[52]	57	—	mostly without diabetes	Dexcom G4 per GlucoBench; —	—	to 303
CITY	R2–R5: training	[62]	153	14–24	T1D; —	Dexcom G5; mixed	2018–19	39/401 codes
DCLP3 (DiaData: DLCP3)	R2–R5: training	[63]	169	14–71	T1D; closed loop vs sensor-augmented pump	Dexcom G6; real-time	2018–19	39/401 codes
REPLACE-BG (DiaData: RBG)	R2–R5: training	[64]	226	≥18	T1D; pump	Dexcom G4 Platinum, 505 software; mixed	2015–16	39/401 codes
SENCE	R2–R5: training	[65]	144	2 to <8	T1D; —	masked Dexcom G4 Platinum run-in, then Dexcom G5; mixed	2017–19	39/401 codes; rare values outside
WISDM	R2–R5: training	[66]	206	≥60	T1D; pump or injections	Dexcom G5; mixed	2017–18	39/401 codes; rare values outside
PEDAP	R2, R3, R5: zero-shot probe; never trained	[18]	103	2 to <6	T1D; closed loop vs pump or injections	Dexcom G6; real-time	2021–22	39/401 codes
RT-CGM (JDRF trial)	R3: training arm (rejected). R5: primary endpoint, 355 subjects with scored windows	[19]	451	≥8	T1D; intensive insulin	DexCom SEVEN, MiniMed Paradigm or FreeStyle Navigator per DiaData audit, per subject not recoverable; wear not stated in the cited report	—	device-dependent; 14–654

`audit_r1.json`), and DiaData records neither the device nor the wear mode behind a subject's data. A re-audit admitted all 451 subjects (modal within-segment interval 300 s and ≥95% of intervals within 270–330 s; `diadata_audit_r2.json`), but that rule could not fail: DiaData rounds timestamps to a five-minute grid [11] and segments split at gaps over 330 s, so every within-segment interval is 300 s by construction. The 96 admitted subjects without a scored window contain no run of 15 consecutive five-minute readings; the other 355 supply the 12.67M windows used in round five. The first audit's collinear fraction, the share of consecutive same-segment reading triples whose middle reading is the exact mean of its neighbours, is 0.305 for RT-CGM against at most 0.229 for the admitted cohorts. The audit does not split that excess between runs of identical readings, which the statistic also counts, and linear interpolation. Runs of identical readings on a five-minute grid could also come from resampling a device with a different native interval (sample and hold), which would distort the rate labels of round five's primary cohort.

Table 12: Model variants bundled in the applications at some point. All use Adam (learning rate 10^{-3}, 20 epochs), the 12-reading window, a five-logit softmax output and static INT8 conversion with the study's histogram observer; every INT8 calibration selection comes from the 27,109 eligible Weinstock validation windows. GAP = global average pooling over time; CE = inverse-frequency class-weighted cross-entropy. Hash prefixes are SHA-256 (float/INT8). The originally bundled artifact (historical protocol, first-200 calibration) has the calibration-study architecture.

Variant	Architecture	Params	Input	Loss	Training data and recipe	INT8 calibration	SHA-256	Status
Calibration-study Trend-CNN, seed 0	CNN (24,36), GAP	2,909	12 absolute readings (mg/dL)	CE	Weinstock eligible training windows (328,575); batch 64, CPU, final epoch	uniform-200, draw 101 (one of the study's 20 selections)	5a0f0744 / 9f2f26f5	historical; study winner
Delta-encoded pair (anchor variant), seed 0	CNN (24,36), GAP	2,909	11 differences + last reading/16	CE	as above	uniform-200, draw 101	c30560f7 / 8590a3d2	historical; champion R1–R2
Round-two winner, seed 1	CNN (96,144), GAP	42,725	as above	CE	DiaData, five cohorts, 898 subjects, 15,135,091 windows; batch 512, MPS, final epoch	uniform-200, draw 101 (cross-cohort)	96eef43c / a887ea4e	historical; champion R3–R5
Round-five winner, seed 4	CNN (96,144), flatten head ($144 \times 12 \to 5$)	50,645	as above	CE	same DiaData windows; batch 512, MPS, cosine schedule, validation-chosen epoch	uniform-200, draw 101 (cross-cohort)	bfb37468 / eacc0a74	bundled now

Table 13: Read ledger: every analysis that read held-out, temporal or RT-CGM labels, in date order (PEDAP, Dubosson and Hall reads included). Dates and short hashes are those of the commit that published each result. Frozen = protocol committed before the read.

Date	Analysis	Labels read	Decision taken
2026-07-01 `6696f5a`	Historical protocol: five-class decision and the earlier revision's scores	Weinstock aggregate test frame (held-out and temporal pooled, before masking)	Five classes kept; scores later superseded as non-comparable
2026-10-10 `9dd88bb`	Calibration study, primary analysis (frozen)	Weinstock held-out, temporal	Uniform-200 calibration adopted; study seed-0 pair bundled
2026-10-10 `9a725c8`	Sensitivity tiers (frozen tiers, run after the primary analysis)	Weinstock held-out, temporal (windows outside the primary mask)	None
2026-10-10 `bdeebf0`, `88655f0`	Second-cohort run (frozen per-cohort protocols)	Dubosson and Hall, held-out and temporal	None; outcomes reported
2026-10-10 `207b7a1`	Delta-encoded inputs (frozen gates; post hoc on previously read data)	Weinstock held-out, temporal	Anchor variant chosen on held-out float recall; bundled
2026-10-11 `aa6127f`	Round one confirmatory read (frozen)	Weinstock held-out, temporal	Gates passed
2026-10-11 `e490ceb`	Round one per-class read of the cached confirmatory predictions (post hoc)	Weinstock held-out, temporal	Not bundled, decided on directional recall; later endpoints changed
2026-10-12 `eff6eac`	Round two confirmatory read (frozen)	Weinstock held-out, temporal; PEDAP	All gates passed; bundled
2026-10-12 `a84b237`	Round three confirmatory read (frozen)	Weinstock held-out, temporal; PEDAP; RT-CGM training labels (rejected arm)	Gate 2 failed; rejected
2026-10-13 `c8f3de6`	Round-four motivation: error structure of round three's saved predictions (descriptive)	Weinstock held-out	Arm choice only
2026-10-14 `cdd26ea`	Round four	None (identity outcome)	Champion retained
2026-10-14 `3755d55`	External review: stable-logit offset scan of cached round-two and round-three predictions (exploratory)	Weinstock held-out, temporal	None
2026-10-15 `3278363`	Round five confirmatory read (frozen)	RT-CGM (primary); Weinstock held-out, temporal and PEDAP (non-gating)	All gates passed on RT-CGM; gates 1 and 2 would fail on both Weinstock components; bundled
2026-10-16 `c739c76`	This revision: descriptive re-analyses of cached predictions	Weinstock held-out, temporal; Dubosson; Hall; RT-CGM; PEDAP	None
2026-10-16 `cca2fda`	Re-review follow-up: round-five RT-CGM contrast without windows holding readings outside 40–400 mg/dL (descriptive, cached predictions)	RT-CGM	None

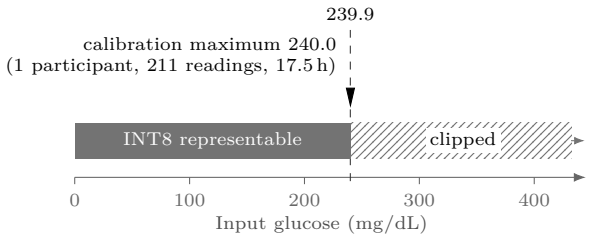


Figure 3: The originally bundled artifact's INT8 input range ends at 239.9 mg/dL, the quantized image of its calibration-set maximum. Above that ceiling lie 18.5% of training readings, and 24.4% of training windows contain at least one such reading (validation: 20.5% and 26.3%). Inputs there are clipped to 239.9 mg/dL.

Table 14: Window provenance and calibration-set composition from the audit. Shares are within each split.

Window source	Windows	Share
Validation, target in validation rows	35,887	64.9%
Validation, copy of a training window	19,402	35.1%
Test, held-out participants	57,068	44.9%
Test, target in test rows	52,831	41.5%
Test, target in validation rows	17,266	13.6%
Test, target in training rows	0	0.0%
INT8 calibration set (first 200 validation windows)		
Participants; segments		1; 1
Distinct input readings; span		211; 17.5 h
Copies of training windows		178 of 200
Maximum input		240.0 mg/dL
INT8 input range		0–239.9 mg/dL

Table 15: Held-out baselines. The CNN row is the five-seed float mean; baselines are deterministic or single fits.

Model	Accuracy	Macro recall	Directional
Constant `stable`	0.727	0.200	0.000
Recent slope	0.662	0.384	0.278
Logistic regression	0.405	0.451	0.461
TrendCNN (float)	0.507	0.494	0.487

C Additional Calibration-Study Details

C.1 Provenance audit

Table 14 gives the window provenance and the composition of the originally bundled calibration set from the audit of Section 7.1, and Figure 3 that set's INT8 input range against the development data.

C.2 Baselines

Table 15 gives the held-out accuracy, macro recall, and directional recall of the baselines and of the float CNN.

C.3 Protocol-to-report map

Table 16 maps each frozen hypothesis and contrast of the calibration study to its outcome.

C.4 Sensitivity tiers

The protocol froze three disjoint window groups outside or inside the primary mask: the exact-observed subset (every checked row coincides with an original reading), windows with gap-filled inputs but gap-free endpoints, and windows with a gap-filled endpoint (whose labels derive from gap-filled grid values and describe the formatter's grid rather than sensor truth). Table 17 evaluates the same frozen artifacts on each tier (summary in Section 7.6); the largest float-to-INT8 gap under uniform calibration is $+1.04$ $[+0.18, +1.86]$ points, on the temporal gap-filled-endpoint tier. The tier evaluation ran after the primary analysis was published, on tiers, artifacts, and contrasts that were all frozen in the protocol beforehand; the full per-model record is in `sensitivity_tiers_r1.json`. A check added in revision removes the 2,264 primary held-out windows whose anchor or label endpoint is a 39 or 401 mg/dL code: the uniform−sequential contrast stays at $+13.27$ $[+10.25, +16.75]$ points (`panel_revision_r1.json`).

C.5 Delta-encoded inputs

Table 18 gives the five-seed results of the delta-encoding experiment (Section 7.8).

C.6 Device benchmark

The prespecified device benchmark (Section 6.1) ran on an iPhone 18 Pro Max (iPhone19,3, iOS 27.0.1), the only study device still available, in 30 paired rounds with the artifact order reshuffled each round (seed 20261010). Per artifact per round, 100 warm-up and 200 timed back-to-back calls run on one frozen synthetic workload, after every artifact is hash-verified and parity-checked on all 277 workload windows; the timer covers input quantization, model execution, and output dequantization, stopping before softmax and class selection, and the thermal state stayed nominal in every round. Model initialization was timed within-process only; peak memory and energy were not measured. Table 19 gives the results; its Δ column is the mean of paired round-level differences against the same round's float mean, and its 95th percentile (p95) is linear interpolation over the 6,000 pooled calls. Raw rounds and the aggregation are in `device_benchmark_ios_r1.json`. Earlier observations of the originally bundled artifacts on a Samsung Galaxy S22 Ultra and the same iPhone are in Appendix A. Later, the delta-encoded, round-two and round-five bundled pairs each passed golden parity on the iPhone with bit-exact INT8 outputs, and single live in-app sessions of 100 inferences per model observed means of 0.080–0.149 ms (round-five pair: float 0.124 ms mean and 0.182 ms p95, INT8 0.126 and 0.179 ms; `MODELS.md`) — observations, not paired-protocol measurements.

Table 16: Protocol-to-report map of the calibration study. Hypotheses are abbreviated from the study plan named in `protocol.json`; contrasts are those of `protocol.json`, whose amendments added uniform versus sequential and the observer arm. Status is judged on the held-out component in points of directional recall; the plan marks secondary intervals descriptive. Untestable as posed = the compared arms shared one input range (Section 7.3).

Frozen item	Outcome	Status	Where
H1: at 200 windows, broader input-range coverage beats random selection with the same participant allocation	Primary contrast +0.06 [−0.02, +0.13]; both arms had the same input range, so the test could not probe range coverage	Untestable as posed	Table 6
H2: the 1,000-versus-200 effect differs between range-aware and random selection	Interactions among the randomized strategies −0.05 to +0.06, every interval including zero; all randomized arms had the same input range	Untestable as posed	Table 6
H3: clipping and rounding explain part of the INT8 change; wider ranges may cost resolution	Clipping reproduces 13.14 of the 13.35-point sequential loss; no resolution cost detected	Partly supported	Table 7
H4: coverage leaves the deployed graph unchanged but raises preparation cost	One size and operator inventory; selection under 0.03 s, export 0.5–0.8 s	Partly supported	Table 19
Primary: range coverage vs. balanced random, 200	See H1	Untestable as posed	Table 6
Range coverage vs. uniform random	+0.01 [−0.05, +0.07] at 200	Descriptive	Table 6
1,000 vs. 200 within each strategy	At most +0.08 for randomized strategies; +13.35 for sequential	Descriptive	Table 6
Budget-by-strategy interaction (no formula frozen)	See H2; uniform vs. sequential −13.33	Descriptive	Table 6
Uniform vs. sequential, each budget (amendment)	+13.38 at 200; +0.05 at 1,000	Descriptive	Table 6
Min–max observer arm (amendment)	At most 0.12 from the histogram observer	Descriptive	Section 7.3
Sensitivity tiers	Uniform beats sequential by +10.5 to +13.2 in every tier	Descriptive	Table 17

Table 17: Held-out directional recall by sensitivity tier: five-seed float mean, sequential-200 INT8 mean, and uniform-200 INT8 mean (seeds × draws); lower block, paired cluster-bootstrap uniform−sequential difference in points. †Post-hoc check on the primary held-out windows without a coded anchor or endpoint ($n = 44{,}112$).

Tier	n	Float	Seq. 200	Unif. 200
Exact-observed	36,150	0.476	0.345	0.477
Gap-filled inputs	8,682	0.450	0.332	0.446
Gap-filled endpoint	2,010	0.501	0.383	0.497
Uniform−sequential, points [95% CI]				
Exact-observed				+13.16 [+10.06, +16.63]
Gap-filled inputs				+11.44 [+6.56, +15.90]
Gap-filled endpoint				+11.34 [+7.14, +15.19]
No coded value†				+13.27 [+10.25, +16.75]

Table 18: Delta-encoded inputs (held-out directional recall, five-seed means; uniform over three draws). The absolute-input rows repeat the main study. Deltas only did not proceed to INT8 (the frozen rule advanced the variant with the higher held-out float recall).

Input encoding	Float	INT8 seq. 200	INT8 unif. 200
Absolute mg/dL	0.487	0.354	0.487
Deltas only	0.497	—	—
Deltas + anchor/16	0.509	0.504	0.509
Uniform−sequential, points [95% CI]			
Absolute mg/dL			+13.38 [+10.29, +16.88]
Deltas + anchor/16			+0.54 [+0.38, +0.72]

C.7 Client packaging, emulator workaround, and parity tests

The SwiftUI iOS application bundles the same two artifact files and synthetic trace by reference to the Android asset directory, so both platforms run byte-identical models, and follows the Kotlin implementation for replay, the gap-resetting window buffer, the model switch, and the latency statistics. Real Android devices use LiteRT 2.1.0 `CompiledModel`; on the tested Apple-Silicon-hosted Android emulator, native model creation hit an illegal-instruction fault in a CPU feature probe for the Arm Scalable Matrix Extension, uncatchable from Kotlin, so emulators are selected before runtime initialization and use the classic interpreter with XNNPACK disabled — a workaround for the tested environment, not a statement about LiteRT configurations generally.

Cross-runtime parity was tested with 20 synthetic golden windows, four selected from each float-predicted class and including values above the INT8 input range. The test also verifies the SHA-256 hash of each bundled artifact. Float logits must match the Python output within 10^{-5}; dequantized INT8 outputs are required to match bit-for-bit because both sides dequantize identical

Table 19: Nine-artifact device benchmark (iPhone 18 Pro Max, iOS 27.0.1, back-to-back calls; inner timer: quantize + run + dequantize, no softmax): the seed-0 float artifact and 8 of the 110 study INT8 artifacts (draw 101 for randomized arms). 6,000 timed calls per artifact over 30 paired rounds; times in μs. Δ is the mean of paired round-level differences vs. the same round's float mean.

Artifact (seed 0)	Mean	Median	p95	Δ vs. float
Float	2.78	2.96	3.17	—
Sequential 200	3.44	3.33	4.00	+0.67
Sequential 1000	3.44	3.33	3.96	+0.66
Uniform 200	3.42	3.33	3.96	+0.64
Uniform 1000	3.47	3.38	4.08	+0.70
Balanced 200	3.45	3.33	4.00	+0.67
Balanced 1000	3.44	3.33	3.96	+0.66
Range coverage 200	3.44	3.33	3.96	+0.66
Range coverage 1000	3.45	3.33	4.00	+0.68

integer outputs with the same float32 operation. The iOS suite applies the same vectors and tolerances and also checks the class-name order. It runs on the physical iPhone, on the iOS simulator, and in continuous integration.

D Additional Sweep Details

D.1 Selection, training, and protocol details

Round three picked its final seed by strict argmax instead of the tie rule, a disclosed deviation; round five applied the tie rule at every pick, including the seed and the kept epoch. The champion's validation references come from its bundled INT8 artifact on the same windows (0.519 directional and 0.563 stable recall for the delta-encoded pair, 0.533 and 0.548 for the round-two pair); stage tables report float checkpoints. Each round's protocol file records the selection trail, candidate, champion SHA-256 hashes, prediction, endpoint, gates, bootstrap seed and pairing rule. For rounds two to five, a pre-sweep smoke run accepted the largest of batch sizes 512, 256 and 128 whose mean validation directional recall stayed within 1 point of batch 64 (512 scored 0.513 and 64 scored 0.506, two seeds each), and two same-seed MPS runs reproduced exactly. The round-five protocol file describes the INT8 calibration pool as Weinstock training windows; the code and the pool hash show the validation pool. An external review published between rounds four and five (`paper/analysis/accuracy-review-2026-10-13/`, commit `3755d55`) scanned stable-logit offsets on cached round-two and round-three held-out and temporal predictions (exploratory; Table 13); its synthetic recency diagnostic and code inspection proposed round five's flatten head and its learning-rate schedule with epoch selection.

Table 20: Round one, held-out per-class metrics, candidate/champion INT8 on the 46,376 paired windows. Recall = sensitivity; precision = positive predictive value. Computed after the pre-registered macro-F1 gates were scored; descriptive.

Class	Recall	Precision	F1
`falling_fast`	0.503/0.572	0.256/0.190	0.339/0.286
`falling`	0.432/0.552	0.274/0.214	0.336/0.308
`stable`	0.688/0.475	0.865/0.911	0.766/0.625
`rising`	0.365/0.334	0.248/0.202	0.295/0.252
`rising_fast`	0.454/0.612	0.342/0.267	0.390/0.372
Directional mean	0.438/0.517		
Macro-F1			0.425/0.368

Table 21: Round four, validation stage means (directional and stable recall, five seeds, float checkpoints on the 27,109 eligible Weinstock validation windows). Stages: H = objective, R = training targets and inputs, W = weight averaging. Control = the round-two recipe (weighted CE, DiaData-only mixture). The best-seed row is the champion's own checkpoint in float; the champion's bundled INT8 reads 0.533 and 0.548 on the same windows. No read was spent.

Stage	Arm	Directional	Stable
H	control (weighted CE)	**0.532**	0.547
H	square-root class weights	0.405	0.827
H	rate regression (Huber)	0.200	0.950
H	CORN ordinal head	0.470	0.543
R	least-squares slope labels	0.531	0.553
R	σ=2 mg/dL input noise	0.495	0.621
W	five-seed weight average	0.238	0.003
W	best single seed (candidate)	**0.539**	0.539

D.2 Round one per-class metrics

A per-class read of round one's cached predictions, made after the gates were scored and not pre-registered (Table 20), shows where the macro-F1 gain came from. F1 rose in every class, but `stable` contributed +14.1 points, half of the summed per-class gain. Precision rose in all four directional classes while recall fell in three of them, and stable recall rose from 0.475 to 0.688: the candidate predicts `stable` more often, consistent with an operating-point shift rather than better separation of the directional classes. Per-class precision and F1: `panel_revision_r1.json`.

D.3 Round four stages

Table 21 lists round four's validation stage means.

D.4 Error profile of the bundled model

Tables 22 and 23 describe the bundled INT8 model's errors on the Weinstock held-out component and on RT-CGM, computed from the cached confirmatory-read predictions. Every movement class is predicted 1.7 to 5.2 times as often as it occurs; `falling_fast`, the most over-called, pairs the highest recall with the lowest precision (0.706 at 0.136 held-out, 0.673 at 0.214 on RT-

Table 22: Error profile of the bundled model (round-five candidate, uniform-200 INT8) on the Weinstock held-out component (46,376 windows) and RT-CGM (12.67M windows). Recall = sensitivity; precision = positive predictive value. Reversal = share of truly falling (or rising) windows labelled in the opposite direction. Anchor bands use the last input reading, with edges from the time-in-range consensus [68], whose two levels below 70 mg/dL are merged here; readings of exactly 180 and 250 mg/dL fall in the upper band. Window counts in parentheses.

	Held-out		RT-CGM	
Class	Recall	Precision	Recall	Precision
`falling_fast`	0.706	0.136	0.673	0.214
`falling`	0.367	0.216	0.494	0.242
`stable`	0.436	0.916	0.605	0.943
`rising`	0.405	0.192	0.569	0.219
`rising_fast`	0.603	0.254	0.483	0.273
Adjacent share of errors	0.640		0.812	
Reversal, falling side	0.138		0.086	
Reversal, rising side	0.151		0.088	
Directional recall by anchor (mg/dL)				
< 70	0.331 (3,038)		0.357 (0.46M)	
70 to <180	0.500 (21,808)		0.549 (8.33M)	
180 to <250	0.522 (11,301)		0.558 (2.75M)	
≥ 250	0.495 (10,229)		0.520 (1.13M)	

CGM), and `rising_fast` has the lowest RT-CGM recall (0.483). Of the windows the model labels with a moving class, 63% on both cohorts have a mean rate within ± 1 mg/dL/min over the next 15 minutes, while a displayed `stable` is right in 92% (held-out) and 94% (RT-CGM) of windows. Most errors fall in an adjacent class (64% and 81%), but 14–15% of truly falling or rising held-out windows are labelled in the opposite direction (9% on RT-CGM). Directional recall is lowest when the last reading is below 70 mg/dL (0.33 held-out, 0.36 RT-CGM; 0.50–0.56 in the other bands), although that band holds only 1 held-out and 96 RT-CGM `falling_fast` windows, of which the model recalls 0 and 2; without that class the ≥ 250 mg/dL band is lowest. Source: `panel_revision_r1.json`.

Table 23: Confusion matrices of the bundled INT8 model, row-normalized (% of each true class); predicted columns in the order `falling_fast`, `falling`, `stable`, `rising`, `rising_fast`. The diagonal is the recall of Table 22. Counts: `panel_revision_r1.json`.

True class	ff	f	s	r	rf
Held-out (46,376 windows)					
`falling_fast`	70.6	15.0	3.9	5.0	5.5
`falling`	35.9	36.7	12.4	9.6	5.4
`stable`	14.4	16.5	43.6	18.4	7.2
`rising`	10.8	6.4	14.0	40.5	28.3
`rising_fast`	7.6	3.4	5.3	23.3	60.3
RT-CGM (12.67M windows)					
`falling_fast`	67.3	17.4	5.4	4.9	5.0
`falling`	27.4	49.4	14.9	5.6	2.6
`stable`	4.2	14.4	60.5	18.3	2.6
`rising`	4.1	4.0	16.3	56.9	18.7
`rising_fast`	6.6	3.9	9.9	31.3	48.3